

Rldcoin: A Peer-to-Peer Payment System Across Delayed Regions

Runlai Deng
dengrunlai@gmail.com
www.rldcoin.com
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Abstract. Rldcoin is a peer-to-peer payment design for communities separated by long or intermittent communication. In the target architecture, every normally started full network node is a relay by default: it discovers reachable neighbors, retains admitted evidence and forwards it over successive contacts, allowing connected communities to extend the network without a central Earth directory. Regional ledgers establish local order, prefunded channels support local payment receipts, and source exports carry value to a separately validating destination. Recognized source finality, authenticated history and a unique import bind the two ledgers; delay-tolerant couriers carry evidence without monetary authority. Conservation depends on valid regional histories, non-conflicting finality, durable records, and uncompromised cryptography. Delivery and spendability additionally require contact, capacity, available signers, and destination progress. This paper defines the target architecture and mandatory acceptance conditions, relates them to distributed-systems and payment research, and specifies tests for delayed routes, archival survival, and protocol evolution. It promises neither immediate remote settlement nor recovery by timeout.

1. Introduction

Bitcoin showed how signatures, public transaction history, and proof of work can order electronic payments without a central transaction processor [1]. Rldcoin begins with the same need to reject double spending, then asks a further question: how can an asset move between regions whose messages may be delayed, interrupted, or carried physically? Delay-tolerant networking research uses persistent store-carry-forward delivery across scheduled or opportunistic contacts [2, 3]. Neither a synchronous global vote across distant regions nor a promise of immediate remote arrival follows from that transport. Treating a sent message as a received payment is unsafe.

The architecture separates local ownership from inter-region delivery. Each Zone has its own chain and validation rules. A source export removes an amount from that Zone's spendable state. A destination may create corresponding spendable value only after verifying an authenticated export and a finalized source checkpoint. A missing receipt or expired delivery deadline cannot authorize a second spend. Earth is the origin region in the reference profile; the same principles apply to more distant Zones.

This is an engineering design and a statement of its trust assumptions, not evidence of an operating interplanetary route. A local channel receipt, source finality, destination import, and a returned receipt are different events. The recipient must distinguish the evidence available at each event. Formal proof-of-work analyses rely on bounded communication and adversary assumptions within a chain [4]; they do not make one globally synchronized chain practical when signal propagation between Zones takes minutes or longer.

Scope and ultimate purpose. The target is one conserved currency that humans can hold, receive and pay through independently verifying local nodes in habitats, spacecraft and distant settlements. With remote regions disconnected, the local economy must continue under its local communication and security assumptions. After physical contact, an owner must be able to transfer to any authorized region, pay there, export onward and return value to an earlier region. Every normally started full network node must contribute a bounded relay service without a separate relay launch or permission from a central operator. As useful contacts and independent resources are added, the network can extend its reach, retain more evidence and offer alternative delivery paths. No always-online Earth directory, Earth approval of each payment or simultaneous global state is required. Peer-to-peer means owner authorization and recipient-verifiable evidence; it does

not remove local consensus or couriers. Local autonomy does not promise safe double spending between mutually isolated recipients with no shared ordering.

A hundred-million-year design horizon. The intended purpose extends to peer-to-peer interstellar payment over humanity's next 100 million years. This is a continuity objective for successive generations of independently adopted protocols, archives and institutions, not a lifetime claim for one key, algorithm, device or immutable software release. Every operational claim must instead name a finite qualified horizon, its assumptions, renewal obligations and failure policy. No known cryptography, post-quantum standard or experiment establishes security for that entire span. Permanent loss of evidence, governance continuity or contact may strand value; the objective does not authorize inflation, a silent reset of ownership or a timeout refund.

Reading conventions. This paper specifies the required Rldcoin protocol. **Must** states a conformance requirement. Sections 5-9 illustrate a proof-of-work and unanimous-checkpoint regional profile; its twelve-block checkpoint count, four signatures and six-successor import maturity are not universal parameters. Each region adopts an exact authenticated profile satisfying the invariants, transition rules and acceptance conditions defined here. Sections 16-21 state the required capabilities, composition assumptions, architecture, risks and qualification conditions.

2. Participants and system invariants

The target network has owners, regional consensus validators, full verifiers, proof couriers and optional channel watchtowers. Owners hold payment keys; independently admitted regional validators order and certify local transitions under Section 18/S3-S4. Verifiers independently execute ledger transitions and validate complete finality/provenance. The Earth reference additionally uses proof-of-work miners and unanimous checkpoint signers for sufficiently buried source blocks; those are reference-profile roles, not the only permitted future architecture. A courier stores and forwards an export proof or receipt across one or more contacts but cannot create value. A watchtower can challenge a stale channel close with a newer jointly signed state. These roles may be held by one operator; separating their names does not establish independent control.

Four invariants govern every transition. Only an owner signature or an expressly authorized protocol rule may move an output. Value can be spendable in one Zone or locked in one journey, never spendable in two places. Accepted state must be reproducible from a pinned genesis and ordered history. Total issued RLD must remain within the cap. Invalid, missing, or contradictory evidence stops the affected transition; elapsed time cannot manufacture authorization.

Communication and fault model. Within a Zone, ledger progress depends on its adopted consensus, connectivity and validation assumptions; mining applies to the proof-of-work reference profile. Between Zones, an adversary may delay, omit, duplicate, reorder, or replay every message, and no finite contact bound is assumed for safety. Signatures and hashes must remain secure for the entire validation horizon. Full verifiers must retain their trusted genesis and finality locks. Eventual delivery is an additional liveness assumption, not a consequence of a valid signature. Compromised checkpoint custody, lost archives, and regional reorganization are separate failures; a secure transport does not repair them.

Safety and availability during a local partition. Remote disconnection does not prevent a sufficiently connected local region from ordering its own payments. A partition inside that region is different: two mutually isolated groups must not both finalize conflicting use of the same inputs. A group lacking the adopted finality quorum must stop new finality and any import or onward export that depends on it, retaining existing balances, requests and evidence. Unfinalized local payments remain subject to the disclosed fork risk. Progress resumes only when the actual local protocol's communication, honest-participation and custody assumptions hold [10, 12, 14]. This design chooses conserved, verifiable settlement over unconditional payment availability; it does not promise both finality and continued spending in every local partition.

For compatible selected histories, let I be distinct issued value, U unspent outputs including immature rewards and imported outputs but excluding value already counted in E or T , E unsettled channel escrow and fee reserves, and T exports debited on the source history but not yet credited on the destination history under review. The accounting objective is $I = U + E + T$, with I bounded by the supply cap. U , E and T are mutually exclusive; transferred fee outputs

remain in U unless explicitly locked in E. Unavailability never deletes a liability. Historical exports and channel receipts are not additional balances. A completed import moves its amount from T to destination outputs, including its fee, even before recipient maturity. This is a conservation argument over compatible histories, not an instantly observable global balance: during disconnection the source cannot know whether T has already become a destination claim. Conflicting finality breaks the compatibility assumption and requires an incident response.

3. Ownership and transactions

The source ledger represents value in unspent outputs. A local transfer names mature, unspent inputs, authorizes them with the owner's signature, and creates recipient and optional change outputs. The input value must cover outputs and a nonnegative fee; execution is atomic. A failed command cannot partially update balances, escrow, exports, or the state root. Signatures establish authority to spend an output, while the shared chain resolves which conflicting spend is accepted.

The smallest accounting unit is the runlai: 1 RLD = 10^{24} runlai. Every amount is an exact integer. Address, transaction, and proof validation bind the applicable chain or Zone identity; an object bound to another genesis has no authority on the selected network. A recipient should verify the selected chain, input maturity, signatures, and confirmation state rather than relying on a broadcast acknowledgment. A signed payment request can bind the recipient, destination Zone, amount, purpose, expiry, and nonce, but the payer still verifies its fields and signs the resulting transfer.

Figure 1 traces two mature outputs owned by one key into a signed Earth transfer. Its amounts are illustrative; the essential checks are network binding, input ownership, maturity, signature validity, non-reuse, and exact conservation. A fee output has reward maturity rather than immediate spendability.

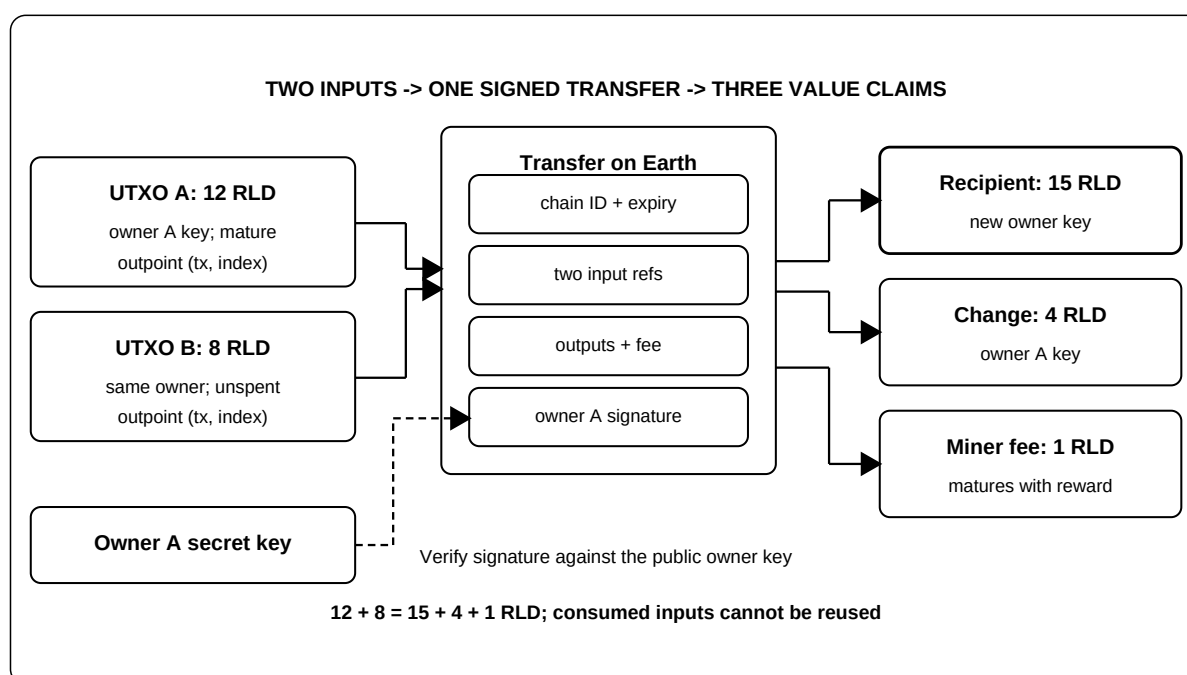


Figure 1. Two input outpoints, one owner signature, and three value claims: $12 + 8 = 15 + 4 + 1$ RLD.

4. Genesis and chain identity

A region's height-zero anchor commits its currency root, regional identity, initial state, target, genesis manifest and exact rule and implementation commitments. The origin begins with zero issued RLD and no initial allocation. Signed, purpose-separated adoption authorizes its consensus and issuance rules; additional regions have zero native issuance and bind the currency root and authorized sources.

Verifiers pin these commitments and authenticate the adopted signer set and rule epochs before replaying history. A chain ID is a network pin, not a substitute for signed adoption and valid state transitions. Test keys and test balances cannot establish monetary authority on a production network.

5. Proof of work and network selection

The Earth proof-of-work reference profile uses double SHA-256 proof of work. A block header binds its parent, chain, height, timestamp, target, miner, commands root, resulting state root, and nonce. The commands root commits the exact ordered command body; it is not a Bitcoin transaction Merkle root. A node validates all commands and state transitions before extending a branch. Subject to an installed source-finality checkpoint, it selects the valid branch with greatest cumulative work. Network messages may arrive out of order; a node can request missing blocks and replay from the pinned genesis.

In the Earth reference profile, the signed birth context fixes the initial target. Difficulty adjusts every 144 blocks toward a 600-second average interval, with bounded adjustment. This is a reference regional parameter set, not a guaranteed block time or a universal interstellar interval. Mining is probabilistic. An unfinalized transaction may be displaced by a competing branch, so API acceptance, block inclusion and finality are separate states.

Figure 2 opens two connected blocks: the header hash links them, while the commands root and state root bind the body and replay result. Proof of work only qualifies a block after the independent command and state checks pass. An installed checkpoint adds a separate branch constraint.

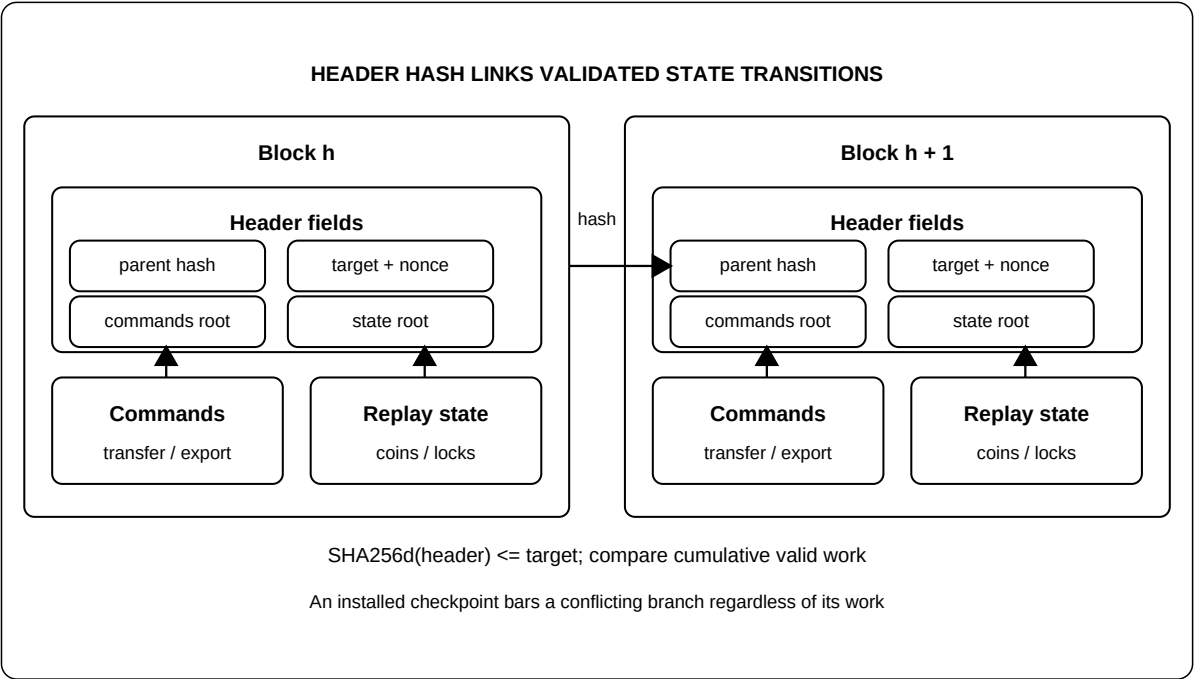


Figure 2. Consecutive Earth blocks bind parent hash, target, nonce, commands root, and replayed state root.

The essential source-block validation algorithm is:

```

ValidateBlock(B, P, C):
  require B.chain == pinned_chain and B.parent == hash(P)
  require SHA256d(B.header) <= B.target
  require B.target == expected_target(P) and B.height == P.height + 1
  require B extends installed checkpoint C
  S = replayed_state(P)
  for command in B.commands:
    S = apply_atomically(command, S) or reject B
  require commands_root(B.commands) == B.commands_root
  require root(S) == B.state_root and supply(S) <= 100 billion RLD
  return valid block with cumulative_work(P) + work(B)

```

A node compares only valid branches that extend its checkpoint; a hash meeting the target does not excuse an invalid command or incorrect state root.

6. Issuance and incentives

The maximum supply is 100,000,000,000 RLD, exactly $C = 10^{35}$ runlai. The origin starts with zero issued value and no initial allocation; additional regions have zero native issuance. The signed origin adoption must bind the following deterministic schedule, its reward beneficiary policy and maturity. An adopted genesis and its issuance rules cannot be retroactively rewritten. The proof-of-work reference profile pays mining rewards and matures rewards/fees after 100 additional source blocks; a different adopted consensus profile must name and qualify its exact beneficiary/finality rule without changing this issuance recurrence.

For height $h \geq 1$, let $L = 200000$, era $j = \text{floor}((h-1)/L)$, and slot $k = (h-1) \bmod L$. At era start R_j is the remaining reserve. If $R_j \geq 2$, budget $B_j = \text{floor}(R_j/2)$; if R_j is 0 or 1, $B_j = R_j$. Let $q_j = \text{floor}(B_j/L)$ and $m_j = B_j \bmod L$. Reward at slot k is $q_j + 1$ runlai if $k < m_j$, otherwise q_j . At the complete era boundary $R_{(j+1)} = R_j - B_j$. All arithmetic is checked integers. The last one-runlai reserve is released once, then every subsequent reward is zero. Genesis height 0 releases nothing. An incomplete era releases only its actually selected blocks' rewards; discarded unfinalized branches do not retain mint. Replayed finalized origin history determines reserve, never elapsed time or a remote report.

```

EraBudget(R):
  require 0 <= R <= 10^35
  return R if R < 2 else floor(R / 2)
Reward(R, k):
  require 0 <= k < 200000
  B = EraBudget(R)
  q, m = divmod(B, 200000)
  return q + (1 if k < m else 0)

```

Boundary vectors: $R_0 = 10^{35}$ gives $B_0 = 5$ times 10^{34} , $q_0 = 250000$ times 10^{24} and $m_0 = 0$, so heights 1 and 200000 each release 250000 RLD. Height 200001 begins era 1 and releases 125000 RLD. For $R = 3$, $B = 1$: slot 0 releases 1 runlai and slots 1-199999 release 0; next reserve is 2. For $R = 2$ and $R = 1$, slot 0 releases 1 and the other slots release 0; subsequent reserves are 1 and 0 respectively. For $R = 0$ all rewards are 0. For $R = 400003$, $B = 200001$, $q = 1$, $m = 1$: slot 0 releases 2 runlai and each remaining slot releases 1. Every complete era releases exactly its budget; no fractional dust or platform-dependent rounding exists. These are normative boundary vectors for profile conformance.

For a valid transfer, input value equals recipient/change outputs plus explicit fees and escrow/export allocations. Source debit precedes destination credit; each export ID creates at most one destination claim. The mutually exclusive accounting buckets in Section 2/18 conserve all issued units, including immature and quarantined liabilities. Receipts and historical exports never create another unit. Only the adopted origin may issue; admission, import, compensation, service fees and copied proofs cannot create reserves. Early issuance can still concentrate ownership. A finite cap does not guarantee purchasing power, liquidity or a sufficient future security/archival budget; Section 18/S14 and R13 require funded service and explicit deficit stops.

7. Prefunded local payment channels

An opening command places a mature source coin into a channel escrow; the baseline additionally requires a separate mature challenge-fee reserve bound to that channel, consistent with the acceptance rule below. A receiver checks both the funded channel and its reservation before accepting a payment state. Each successive state has an increasing sequence number, identifies the same channel, and requires both parties' signatures. The receipt binds that state to an invoice.

A unilateral close exposes its state for 2,016 blocks. During that interval, a higher jointly signed sequence can replace a stale state; an on-chain challenge consumes its reserved fee atomically. Settlement conserves value and returns an unused reservation. Payer, receiver, and watchtower records must survive exact retries and restart. A reliable fast-payment receipt depends on verified funding, data availability, independent observation of stale closes, and tested recovery. It is evidence of a funded channel state, not a claim that an on-chain block has appeared.

Deterministic channel lifecycle. An adopted profile binds its challenge-window count W , fee rule and state encoding; the Earth reference uses $W = 2016$ successors. For close included at height c , deadline $d = c + W$ under checked arithmetic. A valid higher-sequence challenge may be included at heights $c+1$ through d inclusive; settlement is permitted only at $h > d$. Commands are ordered deterministically; a settlement at d refuses even if a challenge shares that block. A challenge updates the accepted state but does not reset this close's deadline. Two different valid jointly signed states at the same sequence are a conflict: retain both and quarantine channel/reserve and their descendants under $S9$, never choose arrival order. An exact same-state retry is idempotent. Funding and reserve remain in E until their atomic transitions; fees consumed move once to their named output/beneficiary.

The reserve must cover the adopted challenge fee when a new receipt is accepted. An owner-authorized mature top-up creates explicit additional reserve without changing channel ownership or already signed allocations. Insufficient/exhausted reserve stops new fast-payment acceptance and cannot pretend a watcher can afford inclusion; fees already spent are not recreated. A close or challenge needing more than the available reserve refuses without partial mutation, unless a separately qualified explicit payer contributes mature inputs. A fixed reserve cannot ensure inclusion under arbitrary fees/censorship. State/journal and highest signed sequence survive restart and qualified independent freshness recovery. A pre-finality close reorganization removes its selected deadline and dependent settlement under the exact replay rule; a new included close gets its own height-bound deadline. Finality locks prevent ordinary rollback after certified settlement; conflicting finals invoke provenance quarantine. Tests must cover $d/d+1$, challenge/settlement order, identical and same-sequence conflicting states, depleted/top-up reserves, rollback, interrupted persistence and stale backup. Leaving watcher coverage requires final local settlement or the disclosed trusted monitoring arrangement.

Locality of the dispute window. A party or watchtower needs to observe a stale close and get the newer state included before the adopted block-height deadline. At the target interval, 2,016 blocks correspond to an expected 14 days, not a guaranteed wall-clock window. Timing attacks on Lightning illustrate why delayed chain observation can defeat timely reaction [15]. This motivates local observers with diverse peers and available challenge fees; it does not transfer Lightning's measured attack probabilities to Rldcoin. A years-distant watchtower cannot protect this channel by responding across the long-distance link. Extending a timeout alone gives no finite guarantee when outages are unbounded. Before leaving local coverage, a user needs a completed settlement or a separately tested, explicitly trusted local monitoring arrangement.

Figure 3 separates the funded off-chain state from the on-chain dispute. The reserved fee is a separate mature coin. A later jointly signed sequence can defeat a stale close during the 2,016-block window, provided the newer state remains available to a party or watchtower.

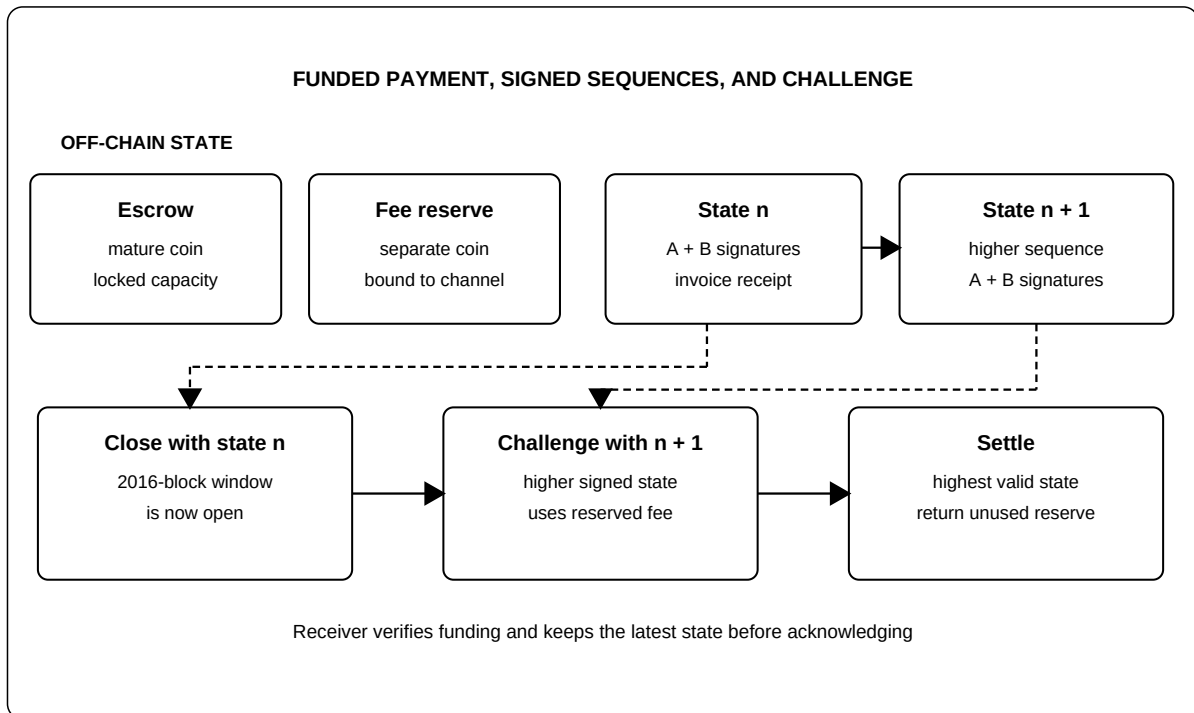


Figure 3. Mature escrow and a separate challenge reserve support signed payment states, stale-close challenge, and final settlement.

```

AcceptChannelState(new, old, funding, reservation):
  require funding is mature and locked to the exact channel
  require reservation is mature and bound to its challenge fee
  require both signatures valid for new.channel and new.sequence
  require new.sequence > old.sequence
  require sum(new.participant_balances) == funding.value
  persist new and its invoice-bound receipt before acknowledging
  
```

8. Export, checkpoint, and destination import

The algorithms and Figure 5 in this section describe the unanimous-checkpoint reference profile. Its local fork and six-block maturity rules do not supply the general onward-finality contract. A qualified region must instead apply its exact adopted local finality and maturity rules, including protection of imported provenance before re-export under I5 and Section 17. A Byzantine-finality profile requires its own authenticated adoption, locking and view-change rules; it is not a reduced-signature interpretation of these algorithms.

An export consumes its source inputs and creates an ordered, unique export record. Before source finality, a source reorganization may remove that transition; such an export is not eligible for adopted destination credit. After finality, the source lock persists, and delivery timeout never refunds it: the destination might already have imported the proof. The export records form a hash tree in sorted export-ID order. A membership proof carries the record, its leaf index and count, and sibling hashes; the verified export root is part of the source commitment and resulting state root. The proof establishes membership in a *claimed* source state, not that the state belongs to the selected source chain or has reached finality. Those checks require replay and a signed checkpoint. Replaying an export ID cannot mint a second destination output.

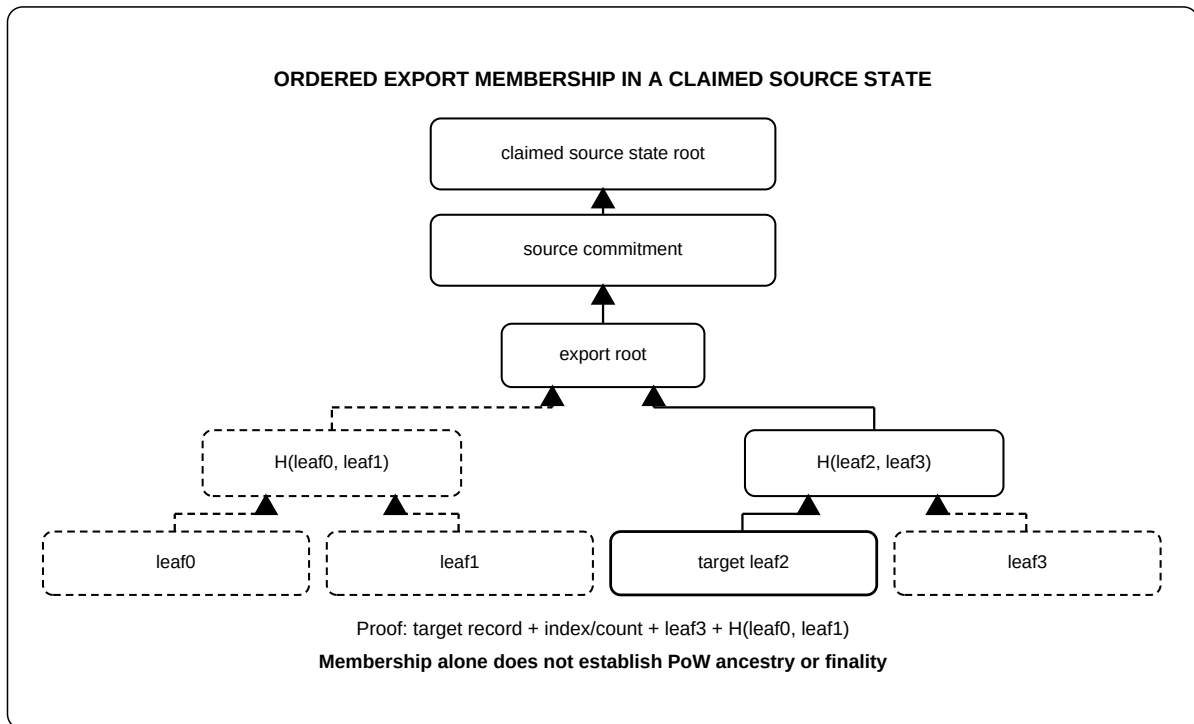


Figure 4. A target export, its index and count, and two sibling hashes recompute the ordered export root; dashed outlines also show surrounding tree context.

A source checkpoint becomes eligible when selected source height H and checkpoint height h satisfy $H - h + 1 \geq 12$: the count includes the checkpoint block, so eleven successors suffice. All four adopted keys sign the same statement, which binds the chain, adoption, block, height, state root, cumulative work, and preceding certificate. Source nodes durably install the certificate and reject any later branch that does not descend from the highest installed checkpoint, even if that branch has more proof of work. The destination checks the certificate against replayed source history and requires six successor blocks after import before the recipient coin can be spent. These two counting conventions differ. Source finality does not finalize destination blocks.

The destination credits each (source chain ID, export ID) at most once on its selected history. A duplicate may return an existing receipt or be rejected as already imported; neither outcome creates a second credit. If a destination reorganization removes the import, the associated outputs and downstream spends must also disappear from that selected state before re-import. An acknowledgment or later spend record may return asynchronously. Loss of that receipt does not unlock the original source output. Clients should expose export, source finality, delivery, import, maturity, and spend separately. A delivery report is not an import receipt, and a returned receipt does not prevent later destination reorganization.

Finality trust and availability. The Earth reference policy uses 4-of-4 unanimity. Four keys controlled by one owner provide no independent fault tolerance. One unavailable or refusing key can stop new checkpoints even while local mining continues. A sufficient condition against two conflicting certificates in the same fixed signer set is an uncompromised signer that checks ancestry and persistently refuses every conflicting statement; validators must also enforce installed checkpoints. This conditional argument is not an independent implementation proof. Rollback, key theft, or inconsistent signer-set changes can invalidate it. A future 3-of-4 policy would need an actual Byzantine agreement protocol with locking, view changes, and reconfiguration; simply lowering the signature count does not inherit HotStuff's guarantees [14].

Figure 5 shows the independent checks on each side of the communication boundary. A courier carries evidence but has no authority to issue value. Finalized source ancestry, matching signatures, unique import ID, and destination maturity each answer a different question.

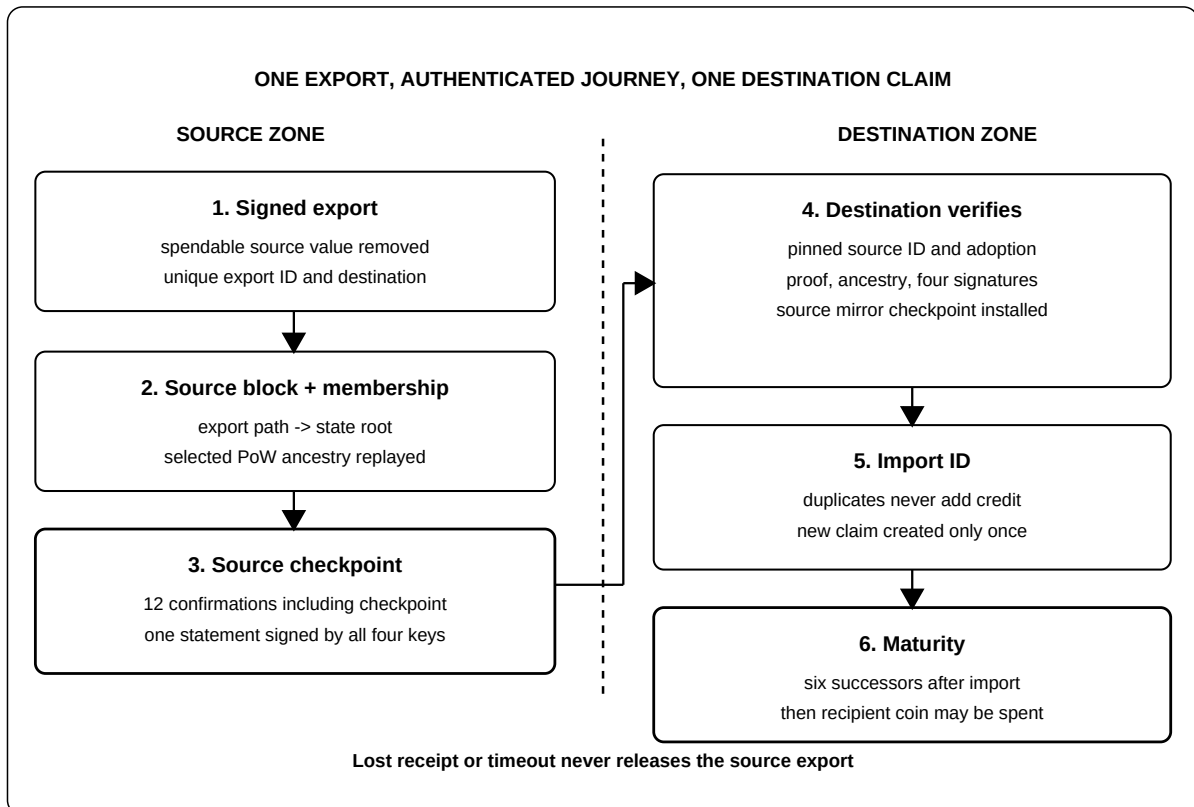


Figure 5. Unanimous-checkpoint reference profile: export membership, four source signatures, unique destination import and six-successor maturity; onward export additionally requires 15 finality.

The checkpoint and import decisions are separate algorithms:

```

SignCheckpoint(block, source_history, signer_lock):
    require block is on selected source ancestry
    require selected_height - block.height + 1 >= 12
    statement = bind(chain, adoption, block, state_root,
                    cumulative_work, signer_lock.previous_id)
    require statement does not conflict with signer's durable lock
    persist the lock before releasing the signature
    return signature(statement)
Import(bundle, certificate, destination_state):
    require bundle.destination == pinned_destination
    require all four signatures on one valid source certificate
    require bundle.export is in the checkpoint's source ancestry
    require membership proof and source replay agree on value and ID
    key = (bundle.source_chain_id, bundle.export_id)
    require key has not already been credited on selected history
    record imported recipient and fee outputs once under key
    allow recipient spend only at import_height + 6 or later
    
```

9. Network operation and delayed communication

Owners broadcast signed commands, miners collect valid commands into blocks, and peers relay blocks or request missing ancestors after reconnecting. During remote disconnection, verified finalized evidence must remain distinct from unknown newer events. Local ledger progress and payments depend on local consensus and security resources; they must not depend on a recent HTTP response from a distant source. A receiving Zone pins the authorized source identity, adoption, proof format and signer epochs. Missing new evidence blocks dependent imports; wrong-domain and unsupported-version messages fail closed. Sending a message does not establish remote completion.

Progressive node discovery and network extension. Every normally started full network node must automatically enter relay service as part of the same node lifecycle: restore its persistent transport identity and queues, discover physically reachable neighbors, exchange authenticated advertisements, learn candidate distant routes, and store and forward admitted evidence for other participants. A separate relay command, central registration or manual configuration of every distant endpoint must not be required. Local adapter, trust and resource configuration may be necessary, but relaying is the default network-node role; a wallet-only client is not a full network node. With no usable contact the relay remains waiting and preserves admitted evidence rather than reporting a connection. The reference topology is Earth ↔ Proxima Centauri ↔ Andromeda: Earth need not connect directly to Andromeda, and no always-reachable Earth directory is required. Discovery begins with a real local neighbor, authenticated seed, scheduled contact or carried advertisement; an entirely isolated node cannot discover an unknown civilization before information arrives. At stellar distances a connection is an asynchronous contact relationship, not a permanent low-latency session. Advertised topology may be obsolete or dishonest, and does not prove current reachability, source freshness or authority to govern a Zone. Forward and return routes may differ or one may not exist [2, 3].

Growth through useful participation. Each node can serve its local community and connect it to further communities through neighboring relays. A newly contacted community can extend the reachable set; an additional usable path can reduce dependence on one courier; independently supplied storage and contact capacity can increase the admitted delivery budget. These are the intended meanings of a stronger network. Node count alone is not a security, throughput or availability guarantee: identities may share one controller, contacts may share one bottleneck, and malicious relays may withhold traffic. Nodes must qualify candidate paths, bound traffic and storage, prevent loops and duplicate forwarding, apply fair admission and retry policies, and attempt available alternatives without discarding valid locked exports. When offered load exceeds service, they must backpressure or refuse new custody before acknowledging it. Relay participation grants no mining, finality or issuance authority [2, 3, 6].

Peer-to-peer payment across the mesh. The payer signs an export for the recipient and exact authorized destination ledger. After source debit and recognized finality, any sequence of relays may carry the authenticated evidence unchanged; an intermediate community need not import the asset. The destination recipient verifies source authority, history, finality and unique import on a local full node, then receives spendable value under local maturity rules. Once those checks pass, local spending does not wait for a return message or Earth approval. Relays cannot redirect the recipient, spend the asset or make a transport acknowledgment count as payment. If the owner instead imports into an intermediate region and later pays onward, that is a new finalized export with the dependency and conservation checks in I4-I7. Missing contact or invalid evidence leaves the relevant payment pending; it never authorizes timeout refund.

A network of local networks. The relay pattern joins communities rather than requiring every participant to speak directly to every other participant. Earth, a deep-space station, a Proxima community, a carrier vessel and a distant galactic community are illustrative local networks. A relay can connect successive networks while each retains its own ledger authority. A mobile carrier may keep evidence until another physical contact becomes available; this does not imply faster-than-light travel. Overlapping circles in the atlas express an adjacent contact relationship, not overlapping radio coverage or a geographic distance scale.

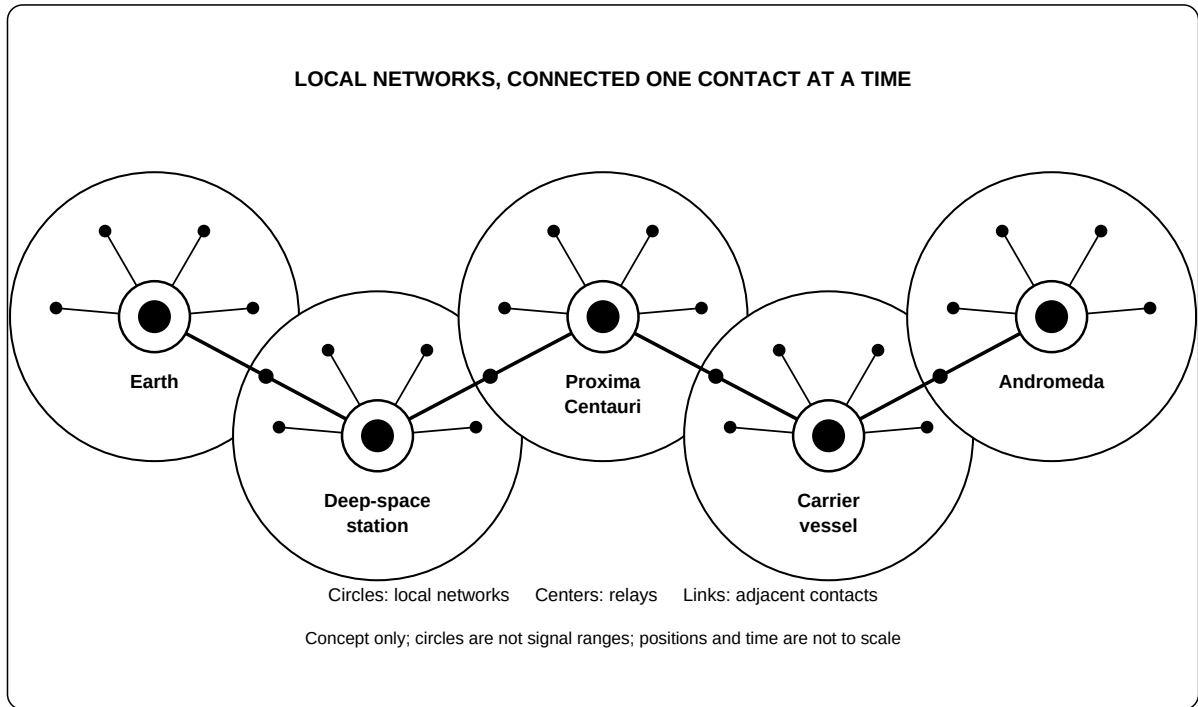


Figure 6. Target architecture: every full network node participates as a relay by default; adjacent contacts join local networks, stations and mobile habitats. Circles are not signal ranges; positions and travel times are not to scale.

Separate discovery, receipt and value. Transport identities and signed advertisements support neighbor authentication and candidate routing; they do not grant monetary authority. Durable queues need explicit limits, retry and custody policies, and capacity refusal must preserve the sender's evidence. A signed transport receipt establishes only the receiver's stated handling of bytes. The ledger separately verifies source authority, finalized history, unique import and maturity. A relay at Proxima may carry an Earth-to-Andromeda frame without importing, reissuing or becoming a consensus signer. Actual onward asset transfer is a separate ledger export requiring the lineage and finality rules in I4-I7.

Galactic scale is a further research boundary. NASA reports approximate distances of 4.24 light-years to Proxima Centauri and 2.5 million light-years to Andromeda [25]. In a simplified stationary-endpoint model, newly generated information takes at least years to the nearest star and millions of years across the galactic example; a relay cannot shorten the sum of physical propagation and waiting delays. Exact moving-endpoint routes need trajectory and time-coordinate models. Supporting a topology on paper does not establish institutional, archival or cryptographic survival for millions of years. Each authorization suite requires independent review and an explicit finite verification horizon; revocation or replacement information also crosses the physical gap. UI states must distinguish learned identity, candidate route, observed contact, queued evidence, destination storage, ledger acceptance and local spend maturity.

A long-distance transfer follows an explicit state machine: locally authorized, source-locked, source-finalized, proof in transit, destination-imported, destination-matured, and receipt returned. Messages may be delayed, reordered, duplicated, or lost. A stable export ID, authenticated ancestry, and deterministic retry behavior make a courier replaceable without letting it mint value. Relays, intermittent networks, and carried media affect availability, not the authority to spend. The recipient can verify spendability on a local destination full node without waiting for a return signal to Earth. Earth can verify that outcome only after destination history and a receipt reach an Earth-side verifier. A timeout can trigger investigation or a second courier, but it cannot prove the destination did not import. This distinction resembles asynchronous cross-ledger packet protocols [5], although Rldcoin's adopted source-finality and no-silent-refund rules are its own.

For one complete evidence bundle carried sequentially through hops i , let L_i be one-way propagation distance divided by signal speed, W_i the actual non-overlapping contact wait after arrival at that hop, and X_i its serialization and relay time. Let T_{source} include export inclusion and source finality, and T_{dest} include destination verification, import inclusion, and the six-block spend maturity. A lower-bound decomposition for this store-and-forward model is:

$$T_{\text{remote_spend}} \geq T_{\text{source}} + T_{\text{route}} + T_{\text{dest}}$$

$$T_{\text{route}} \geq \sum \text{over hops } i \text{ of } (L_i + W_i + X_i)$$

$$T_{\text{maturity_receipt}} \geq T_{\text{remote_spend}} + T_{\text{return_route}}$$

These are durations, not promises of arrival. A scheduled contact plan can estimate W_i and capacity; an unscheduled or failed contact may make it unbounded [2, 6]. Mars links illustrate that even one-way light time varies with geometry and communications can be disrupted [7]. The return path may be different or absent. A local funded-channel receipt can be fast because its parties share a local validation context; it cannot certify remote destination maturity before evidence returns.

Interplanetary versus interstellar scale. NASA's representative Mars mission profiles include roughly 21-22 minutes of maximum one-way delay and blackout examples of about 13 or 21 days [7]. These are scenario-specific, not route guarantees. Proxima Centauri is about 4.25 light-years away [24]; under a simplified stationary-endpoint model, new evidence needs at least about 4.25 years one way and an immediate reply at least about 8.5 years round trip, before other delays. Moving spacecraft require a specified trajectory and time coordinate instead of this approximation. Relays cannot shorten the causal light-time bound. A destination liquidity provider could pay from funds already present locally, but would extend separately priced credit against an unconfirmed journey; that is a proposed service with counterparty and inventory risk, not faster arrival of the exported asset.

An export proof is an application payload for a persistent queue. Each relay should bind payload bytes to the export ID and intended destination, authenticate its peer or bundle where appropriate, retain the payload until durable custody transfer or a documented retention limit, and record receipt, forwarding, and deletion events. Contact scheduling must account for payload size, link capacity, competing traffic, and storage exhaustion [2, 3, 6]. Bundle Protocol lifetime expiration or a courier's deletion only ends that copy's delivery attempt; it does not reverse the ledger export. With imperfect clocks, transport lifetime and ledger validity must not be inferred from one another [2]. Bundle-layer integrity or confidentiality can protect payloads in transit but cannot substitute for source replay, checkpoint signatures, or the destination's unique-ID check [8].

Long-delay transport adaptation. A long-delay adapter must support asynchronous authenticated evidence and independently measured contact/custody boundaries without requiring an immediate reply, a continuously open session or universally synchronized clocks. It must bind versioned identity and replay domains, retain complete evidence across failed attempts, specify authentication when a key or certificate leaves its qualified horizon, and separate transport failure from monetary refund. Interactive pinned-TLS connections may serve connected local links, but their connection challenges, socket deadlines and certificate policies do not establish suitability for a years-long contact. BPv7 carriage, link authentication and durable custody require an explicit adapter profile and route-specific qualification [2, 8].

BPv7 does not itself supply an end-to-end monetary acknowledgment or a mandatory durable-custody protocol; RFC 9171 moved custody transfer out of the base protocol. An operational custody profile therefore needs its own authenticated retention and deletion obligations. BPSec protects selected bundle data through integrity and confidentiality services; RFC 9172 explicitly does not provide hop-by-hop authentication as a service [8]. Neither standard proves that a remote ledger accepted value. Relay signatures make statements attributable but do not, by themselves, prove correct clocks, truthful delivery, or physical location. Transport lifetime, invoice expiry before signing, and ledger spend-height rules must remain separate.

An application-layer reference carriage profile uses the following bounded frame format for source and destination evidence. A versioned canonical JSON frame contains one of five kinds: source sync page, source finality certificate, finalized import command, destination sync page, or destination receipt. It binds both 32-byte chain IDs, the 32-byte export ID, the exact payload bytes, their SHA-256 digest, and a domain-separated message ID. Each payload is at most 3 MiB; a durable queue accepts at most 4,096 frames or 256 MiB and fails closed at capacity. Repeated receipt of identical bytes is idempotent, while carrying a frame leaves the original queue copy intact. These limits are transport controls, not consensus limits. The message ID detects changed bytes but is not a signature or proof of sender identity. Operators must authenticate peers or use bundle security for an actual route [8]. Frames can be carried by files, contact media or a separately qualified BPv7 adapter. This frame format does not itself specify contact scheduling, independent custody or a physical link [2, 6].

In the unanimous-checkpoint reference profile, the receiver replays source blocks from its pinned source anchor, installs a valid four-signature certificate on its local source copy, then submits the finalized import to a destination node that independently mirrors that source. The destination node validates the proof and unique ID before local inclusion. To report the result back, a courier must return selected destination blocks and an import receipt; an Earth-side destination mirror must replay those blocks against the same source identity and verify the receipt on its selected branch at or above the receipt's recipient spendable height. The adopted rule adds six successor blocks after import, so an import in block 91 becomes spendable at height 97; six confirmations counted *including* block 91 reach only height 96 and are insufficient. A receipt alone is an index into history, not an independent finality proof. A source-region verifier must not interpret its own stale destination mirror as evidence of non-import.

```

Frame(kind, source, destination, export_id, exact_payload):
  require pinned source != pinned destination and payload <= 3 MiB
  body = canonical_json(version, kind, source, destination,
                        export_id, SHA256(exact_payload), base64(exact_payload))
  message_id = SHA256(ASCII("RLD-INTERREGION-EVIDENCE-V1") || 0x00 || body)
  persist frame and message_id before a contact copies exact bytes
  return message_id # integrity and deduplication, never value authority
ReceiveAndSettleReference(frames, local_source, local_destination):
  require each frame's route, digest, and message_id match pinned IDs
  replay ordered source pages and verify every block and state root
  install source certificate only if signatures and ancestry verify
  submit finalized import; require destination independently accepts it
  observe selected import plus six-block maturity on destination
  return destination history and receipt; replay on Earth side
  require selected destination height >= receipt.recipient_spendable_height
  never infer import, non-import, or refund from contact silence

```

Figure 7 makes the epistemic limit explicit. After the source finalizes an export, silence is consistent with both a proof that never reached the destination and a completed import whose receipt was lost. Those worlds are indistinguishable at the source without new authenticated evidence. A source-only time-based refund would release spendable value in the second world and violate conservation. Atomic-swap timelocks can work under their own bounded-time and observation assumptions [9]; they do not make an unverified absence of import a proof in this model.

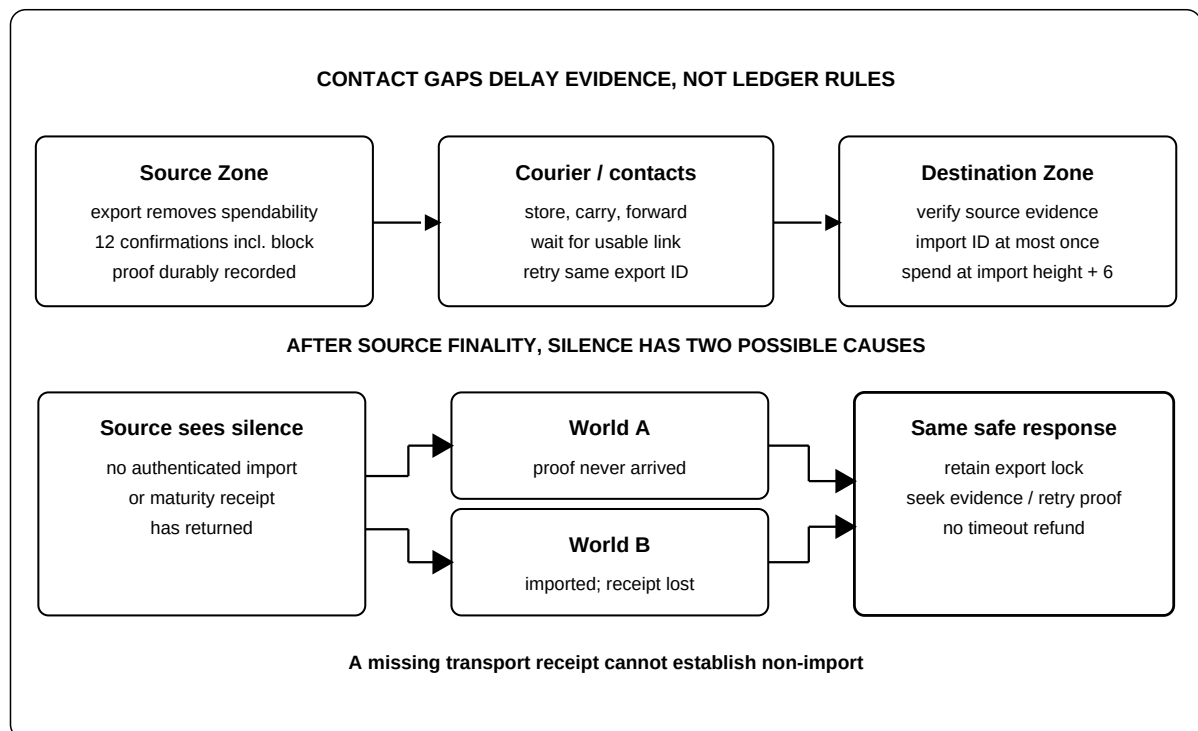


Figure 7. Store-carry-forward proof delivery and two indistinguishable silent outcomes; neither a contact gap nor a missing return receipt authorizes a refund.

```

RelayExport(proof, export_id, destination):
    require proof bytes and destination match the queued export ID
    persist payload, hash, and forwarding state before acknowledging custody
    for each usable contact within the adopted custody obligation:
        send identical payload; record peer receipt and retry safely
    report delivery evidence separately from destination import evidence
    reclaim courier copies only under Section 18/S13 custody rules
    never issue or refund value as a consequence of transport status

```

An operating route requires a pinned destination, independent operators and key custody, tested contacts, durable capacity, and disconnection, restart, reorganization and contradictory-certificate qualification. Safety does not imply liveness: without contact, storage, available verifiers and non-conflicting finality, there is no finite bound on remote spend. Fully asynchronous deterministic consensus cannot guarantee termination in every execution under the crash model of FLP [10]; the design therefore states conditional regional progress rather than unconditional global consensus.

10. Verification and storage

A full verifier pins the exact genesis and adopted rules, checks signatures and block work, replays commands in order, and compares resulting state roots. Startup and replay reject wrong identities, corrupt durable records, unsupported history, or noncanonical inputs. Source and destination can be checked from their public records and published release rather than from a website badge. A lightweight client that trusts reported roots or a remote API inherits that operator's availability and honesty assumptions.

Old blocks and proofs may be indexed or compacted for convenience only if the commitments and sufficient data remain available for independent replay, destination import, channel challenge, and dispute resolution. An index accelerates lookup but is not an authority independent of the committed history. A light client that accepts a header and membership path without executing every command relies on full verifiers to detect invalid state and on its chosen source of the best chain. That trust assumption must be visible to the user. For a delayed route, archival policy must outlive the longest intended journey and possible dispute, including the source ancestry, export membership path, signed checkpoint, adoption records, import record, and receipts. A hash commitment cannot reconstruct data that every custodian has deleted. Independent archival copies and tested recovery are therefore part of availability, not a relaxation of the verification rule.

The baseline requires complete authenticated source replay. An adopted compact or validity-proof profile must bind the exact genesis, rule version, source state, finality policy, destination, amount and consumed export ID, and preserve provenance, incident detection and recovery semantics. Data-availability research separates commitments from availability and gives light-client guarantees under additional network and sampling assumptions [20]. A disconnected verifier cannot treat a missing fraud proof as evidence of validity. Proof compression, erasure-coded recovery and availability sampling cannot substitute for those obligations without an authenticated profile and independent evidence. Archives must preserve decoding software and versioned validation rules as well as bytes.

11. Keys, privacy, and user interfaces

Owner keys authorize payments; genesis and checkpoint keys sign only their narrowly defined network statements. These roles need separate custody, backups, and purpose-separated signing domains. A wallet should display the actual genesis, recipient, amount, fee, destination, expiry, and state before signing. A payment request can help bind these terms but cannot replace transaction validation. Secret keys should never be sent to a website, courier, or mining peer.

User payment contract. The payer selects a locally trusted currency and region, checks the recipient's exact destination genesis and adoption evidence, reviews amount, fees and input ownership, and authorizes a new local debit. Before inclusion, a queued request is not a debit; after recognized source finality, the wallet must show the asset as committed to that destination even during permanent separation. The recipient uses its own locally adopted verifier to distinguish unknown or incomplete evidence, verified evidence pending import, imported but immature value, locally spendable value and eligibility for a new finalized export. A source wallet separately shows whether destination evidence has returned; lack of that knowledge cannot negate the recipient's accepted state or release the source debit. Local invoice

expiry, signing-height validity and transport retention are distinct. Recovery and signing require current local qualification, rather than a stale API display.

Cryptographic authorization and confidentiality. Each adopted profile must identify its exact signature, hash, key-establishment and storage-encryption suites, parameters, canonical encodings, purpose domains, key roles, activation/retirement rules and finite verification horizons. Payment, channel, finality, admission, governance, discovery and custody statements require independently reviewed authorization appropriate to their roles. Ed25519 illustrates a classical discrete-logarithm signature assumption; exposing a public key exposes that assumption to the attack considered below. Encrypting stored keys protects custody under its password, device and implementation assumptions, but cannot strengthen the public authorization algorithm. Transport must authenticate peers and bind the exact negotiated key-establishment policy; a TLS version or certificate pin alone cannot establish that exchange or its post-quantum security [33, 34]. Required long-term operation includes independently qualified post-quantum authorization, confidential transport and authenticated migration under I11.

Quantum attack boundary. A sufficiently capable fault-tolerant quantum computer could apply Shor's discrete-logarithm algorithm to derive an Ed25519 signing secret from its public key [30]. That threatens owner debits and channel closes, forged validator quorum/finality evidence, regional admissions, upgrades and relay identity. A three-of-four quorum is not protection when the attacker can forge those exposed keys. Hiding a future public key behind a commitment can postpone exposure in some designs; it cannot rescue directly exposed addresses, validator keys or a key after disclosure. This is a capability-based future threat, not evidence that today's machines have broken these Rldcoin keys. There is no reliable attack date on which to base a years-long journey.

Shor's signature attack does not also invert SHA-256. Grover's idealized search gives a quadratic speedup for preimage search and proof-of-work target search; collision attacks have a different quantum query bound [31, 32]. Query bounds omit hardware, error correction, depth, memory and parallelism constraints. They are not a forecast of practical mining advantage, and a collision is not an arbitrary valid history rewrite. Double hashing is not a general quantum-security multiplier. Each adopted hash suite must state its preimage, second-preimage and collision requirements; work economics need a separate quantum-adversary analysis. A wider hash, such as SHA-384 or SHA-512, is an evaluation candidate requiring explicit migration of commitments and identifiers, not a drop-in change. Symmetric encryption, password entropy, nonce uniqueness, randomness and side channels likewise need their own budgets.

Post-quantum candidates and route costs. NIST finalized ML-DSA and SLH-DSA signature standards in 2024 [11, 21]. They are candidates for future owner, consensus, checkpoint and governance signatures, not guarantees against all future cryptanalysis. ML-KEM, standardized in FIPS 203, establishes a shared secret for a reviewed encrypted-transport construction; it is not a transaction signature and does not authenticate a peer on its own [26, 27]. Recorded classical key-exchange traffic may face later decryption even after a deployment upgrades, so confidentiality horizons must be planned before sending sensitive evidence. Ledger signature migration and transport key establishment must both be qualified. RFC 10024 standardizes X25519MLKEM768, SecP256r1MLKEM768 and SecP384r1MLKEM1024 hybrid exchanges for TLS 1.3 [35]. They are candidates for connected local/ground links, with separately qualified peer authentication; they neither replace payment signatures nor establish an asynchronous stellar adapter.

Parameter selection must account for relay fragmentation, queue admission, verification cost, device custody and archival replication. ML-DSA-87 has a 2,592-byte public key and a 4,627-byte signature; the SLH-DSA SHA2/SHAKE 256s profiles have 64-byte public keys and 29,792-byte signatures, while 256f signatures are 49,856 bytes [11, 21]. Three separate signatures alone would consume 13,881 or 89,376 bytes respectively, before public keys, certificates, history and carriage overhead. Those arithmetic examples are not an adopted quorum format or an aggregation scheme. Lower-frequency archival/governance checkpoints may justify algorithm diversity, but only after security and bandwidth review. Limits cannot be silently increased or evidence deleted to fit a larger signature. Every adopted suite requires measurements and malicious-size tests; no new wire-size or security claim follows from a standard's publication.

Cryptographic agility and migration contract. Every future authorization and evidence format must bind a registered suite and exact parameter set, canonical encoding, purpose/domain, currency and regional genesis, protocol version, key role and epoch, policy/activation commitment, signer identity and full intended statement. An upgrade must define supported suites and cryptographic horizons, including hash and transport suites, and reject unknown, ambiguous,

mixed-domain or downgraded evidence. Algorithm choice is authenticated policy, not peer-controlled negotiation. Hashes and addresses require versioned namespaces; a new digest or public-key encoding must not reinterpret an old export ID, output or consumed-import record. Domain separation alone does not establish this complete agility contract [28].

A proposed transition first inventories exposed owner, validator, archive and governance keys, then qualifies implementations and publishes reproducible vectors and review. While old authority remains trustworthy, an authenticated transition must bind the exact predecessor state, new suites and keys, proof of possession, approval roles, activation gates, preserved locks and rollback prohibition. During a declared overlap, a reviewed hybrid signature policy may require both classical and post-quantum verification of the same intent; accepting either signature would retain the weaker authorization path. A custom combination is not assumed secure: its transcript, key binding, failure behavior and composition need review. RFC 9955 discusses hybrid-signature terminology and stripping/downgrade concerns; it does not standardize a specific Rldcoin combination [36]. After the adopted retirement gate, new authorizations must meet the current policy. Owners must migrate spendable balances and channel authority while the prior authorization remains trustworthy. Dormant balances that miss the gate require a declared freeze/recovery policy; a forged-capable old signature alone cannot authorize a new key or establish ownership. Historical replay remains a separately specified function and cannot grant fresh value authority to an obsolete signature. Transport hybrids also need an established reviewed construction. RFC 10024 depends on the TLS 1.3 construction and transcript; copying its KEM/key-share components into an asynchronous carrier does not inherit TLS security. Historical replay of retired suites is distinct from fresh debit/migration authority: self-declared old timestamps cannot prove a signature predates key compromise. Old-key forged migration chains, conflicting renewals, dormant balances and nodes returning with an entirely old view must be rejected or kept unavailable unless independently preserved trust/renewal evidence establishes the adopted policy. These migrations require independent adversarial qualification.

Public keys are pseudonyms, not anonymity. On-chain amounts, timing, and output relationships are visible; reused keys and multi-input transfers can reveal further links. New keys can reduce casual linkage, while cross-Zone proofs necessarily reveal enough history to justify an import. Privacy must account for the ledger and network metadata rather than assume that a cryptographic address hides identity.

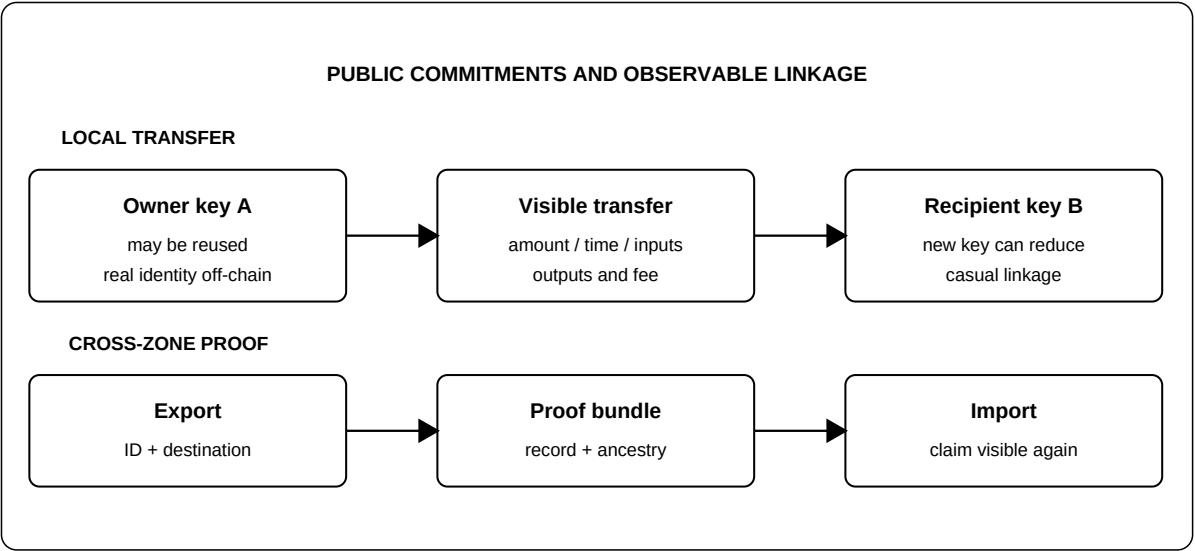


Figure 8. Public local transfers and cross-Zone proofs expose different links; a fresh key reduces simple reuse but does not hide value or timing.

12. Recovery and protocol evolution

Nodes recover by authenticating the original genesis, durable history, installed checkpoints, and executed state before serving value. Recovery must retain per-key signer locks and the most recent signed channel states. Restoring a checkpoint key without its last signed statement can permit equivocation; restoring a channel without its newest state can permit a stale close. A restart cannot turn an in-transit export into a refund or a pending receipt into a settled balance.

An upgrade requires an explicit version, activation rule, exact rule and code commitments, replayable migration, and preserved asset history. A software release alone cannot change another verifier's accepted genesis or supply. New Zones require their own signed identity and source bindings. Unknown formats must fail closed instead of falling back to an older interpretation. A public rule change must state how prior signatures, outputs, channels, exports, imports, and checkpoints remain valid or are resolved.

Admission, epochs and governance. Removing an online Earth directory does not remove trust: every verifier starts from its adopted currency root, admission authority and regional finality policy. New regions, validator sets and rule changes must carry an authenticated, ordered authorization chain from that context; route discovery or a software download cannot create it. A transition must bind the exact closing history and old/new approval roles, preserve all earlier finality and duplicate-import obligations, and maintain separately retained signing/caller locks for continuing and new validators. Missing quorum, stale or incomplete transition evidence, or an unknown policy stops the dependent transition; operators may not reset a lock, lower a threshold or use a new genesis to resolve existing value. An isolated verifier cannot know an undelivered revocation or conflict. Its versioned acceptance policy must therefore state its trust and cryptographic horizon and preserve unsupported evidence without accepting new value from it. Regional authorization and incident-resolution governance must have a reviewed specification, independently held authority and adversarial qualification; a same-controller handoff sample cannot establish independent operation.

Disconnected upgrades and trust anchors. Regions cannot activate a change simultaneously across stellar light time. Before accepting an in-flight export, a destination must establish the original authorization policy, complete transition and renewal ancestry, retained spent/export identifiers and its own locally adopted verification horizon. If a journey or disconnection can exceed that horizon without timely authenticated renewal, the route is not qualified for newly accepted value. Unsupported or expired evidence stays preserved and pending; silence neither validates it nor refunds the source. Revocations bind a key, role, epoch and compromise scope, but cannot protect a destination before authenticated notice arrives. Key expiration is not proof of non-compromise. Recovery from an old backup must detect missing trust-anchor updates and monotonic locks before serving value. The backup alone cannot establish that it is latest: Section 18/S11 requires a surviving qualified independent freshness/monotonic witness, and missing evidence forbids restored signing or spending.

An independently held trust anchor is needed to distinguish genuine transition history from a forgery once an old key is broken. Multiple independently operated custodians, authenticated update chains and preserved contradictory records reduce dependence on one institution; they do not prove eternal continuity. If valid competing governance branches exist, the protocol must expose the conflict and quarantine dependent acceptance until an explicitly adopted resolution establishes a trusted branch. A new genesis cannot erase prior liabilities. Permanent partition may leave distinct local adoption histories without a globally agreed upgrade; no cryptographic rule forces reconnection or restores shared authority after it is lost.

Evidence renewal and generational recovery. Archives must retain original signed bytes, replay rules, authorization/epoch ancestry, dependency and consumed-export records, decoding tools and independent recoverable copies. Before the protecting signatures or hashes lose their accepted strength, independently authenticated renewal must bind the complete still-valid evidence and policy into new commitments; hash renewal requires access to the protected bytes, not just re-signing an obsolete digest. Renewal records preserve original history and their own ordering/trust evidence. Evidence-record methods offer a preservation model, not a Rldcoin consensus authority or a guarantee that a remote timestamp is honest [29]. Re-signing after original authenticity is lost cannot identify the genuine conflicting past. Physical media refresh, diversity, repair and funded custody must continue across generations, with tested loss and recovery.

Key loss needs pre-adopted recovery authority, scopes, thresholds and anti-replay rules while ownership can still be authenticated. Without a valid surviving authorization chain, an operator cannot mint replacement value, transfer an orphaned balance or choose a claimant from an unsupported old signature. Losing all copies, all qualified keys or all governance continuity can make assets unavailable permanently. Neither a post-quantum algorithm nor archive renewal prevents that failure. The 100-million-year objective therefore requires ongoing reassessment, finite-horizon route admission, institutional succession and explicit failure states; it remains an unproved continuity program.

For a stranded export, recovery starts by querying independently held source and destination evidence and replaying both histories. If the destination has imported, resend or reconstruct the receipt. If import is not observed, keep the export pending and retry the same destination-bound proof through another courier. Absence from an incomplete destination view is not non-import proof. Cancellation is excluded from the baseline. An explicitly adopted extension must finalize a destination transition that both rejects an existing import and permanently consumes the export ID as cancelled, including against delayed copies and replay after recovery. The source must verify that transition under an adopted destination-finality rule before unlocking. A historical non-membership proof alone cannot prevent a later import. Such cancellation and its required destination finality do not exist in the current protocol; they would add communication and governance assumptions, not guarantee recovery from permanent separation.

13. Security analysis and economic recovery

An attacker may double-spend an unfinalized local transfer by outworking the observed branch, censor or withhold a contact, replay an export, substitute another Zone's proof, exhaust relay storage, present a stale channel state, steal a key, or cause signers to authorize conflicting checkpoints. Work, signature, domain, maturity, unique-ID, and state-root checks address different parts of this model. Proof of work is probabilistic; channel safety needs available evidence during its challenge window; source checkpoint safety requires adopted signers not to authorize conflicting histories and all value-serving source nodes to enforce the lock. Redundant couriers and contact paths can improve delivery odds but do not prevent signer equivocation or invalid destination validation.

For an idealized proof-of-work race, let p be the honest share of work, q the attacking share, and z a current block deficit. With independent block discoveries, the probability that the attacker ever catches up is $(q/p)^z$ when $p > q$, and 1 when $p \leq q$ [1]. This is a catch-up model from an observed deficit, not an end-to-end payment guarantee. It excludes network partition, signer equivocation, channel failure, and destination reorganization. Those conditions must be evaluated separately.

If z instead counts honest confirmations observed while an attacker mines privately, the attacker may already have found k blocks. Under the idealized independent-discovery and Poisson approximation used in [1], let $\lambda = z \cdot q/p$. For $q < p$, the approximate eventual catch-up probability is:

$$P(z, q) = 1 - \sum_{k=0}^z \left[\exp(-\lambda) \cdot \lambda^k / k! \cdot (1 - (q/p)^{(z-k)}) \right]$$

$$\lambda = z \cdot q/p, \quad p = 1-q, \quad q < p$$

Illustrative values, rounded from that model, are:

Honest confirmations z	$q = 0.10$	$q = 0.30$
1	0.204587	0.627749
3	0.013172	0.324584
6	0.000243	0.132111
12	0.000000089	0.023584

Figure 9 uses a logarithmic axis to display that decline. Neither this Poisson calculation nor its illustrative inputs measure Rldcoin's live hashrate or guarantee an import, because the source checkpoint's signer custody and the destination chain add separate risks.

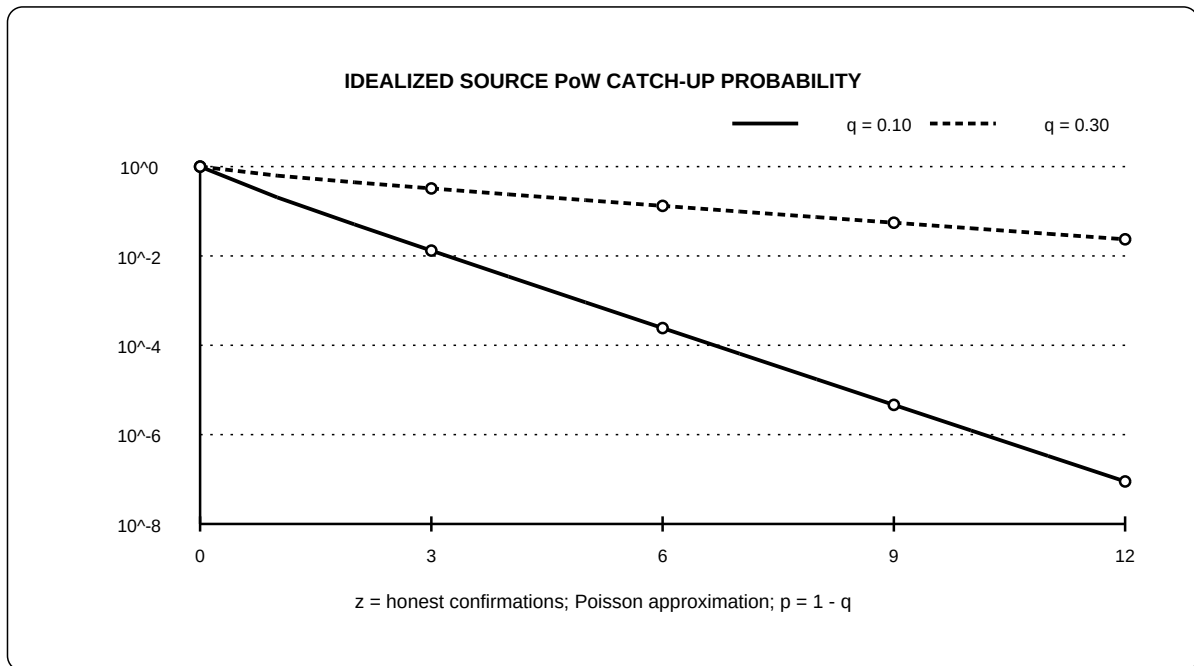


Figure 9. Idealized source proof-of-work catch-up probability after z confirmations, for illustrative attacker work shares of 10% and 30%.

The protocol can reject an ordinary source reorganization after a unanimous checkpoint, but cannot automatically restore value already spent at the destination after signer equivocation or compromise. If contradictory certificates appear, affected imports and redemptions should stop, both certificates and liabilities should be published, and any compensation should follow an auditable rule. An external value service needs a cap on outstanding credit, matured source collateral, an incident ledger, and a reproducible allocation process. A shortfall must be disclosed rather than covered by silent extra issuance.

Conflict recovery has a dependency scope. A valid contradictory finality statement invalidates the non-conflicting-history assumption for the affected region and dependent imports. Verifiers that learn it must retain both signed histories, quarantine affected existing balances and all descendants through local payments, fees, channel settlement, splits and merges; stop every new value transition consuming that lineage, including local spends, imports and onward exports, and disclose previously paid exposure; unrelated regions do not become a globally synchronized emergency quorum. Evidence must propagate through later contacts, so a disconnected recipient may already have spent before learning of the incident. Restoring archives cannot make an authentic conflict disappear or guarantee clawback. Resumption requires the explicitly adopted incident-resolution authority and evidence, with no hidden issuance. Relay silence, an unsigned allegation or an operator status badge cannot choose a winning history.

Independent miners and operators, separate key custody, outside review, capacity testing, and long-running recovery improve assurance. Four signatures controlled by one person do not establish independent governance. An operator status page helps users observe a service but cannot prove custody, consensus safety, or availability of a distant route. Each new Zone and route needs evidence for delay, disconnection, conflicting proofs, and operator failure. No protocol can make a message arrive before physical communication permits.

Destination security has its own budget. A destination region has zero native issuance; source mining rewards do not automatically pay its miners or provide its hashrate. Import/transfer fees, operator funding, concentrated hash power, and destination reorganizations must be evaluated separately. A fixed global cap is not a security-budget proof, a guarantee of purchasing power, or a promise of liquidity. Long-haul transport and archival fees also need explicit payers; unpaid preservation cannot be assumed for decades.

14. Related work and design choices

Consensus and asset transfer. Partial-synchrony research explains how termination can depend on eventual communication bounds [12], while FLP excludes unconditional deterministic consensus termination under its asynchronous crash model [10]. It does not prove that every payment needs global consensus. Guerraoui et al. show that single-owner asset transfer has consensus number one and give a Byzantine message-passing construction using secure broadcast [13]. That remains a communication and fault-model result. Applying it to Rldcoin's mining, shared channel states, checkpoints, and reconfiguration would need a new composition proof. The target therefore preserves adopted regional ordering while treating causal asset-transfer designs as an explicit research alternative.

Partitioned ledgers and cross-chain evidence. Monoxide studies asynchronous consensus Zones for blockchain throughput [23]; its terrestrial sharding results do not establish years-long partition tolerance. IBC separates authenticated client state and packet processing [5], and atomic-swap protocols state assumptions for timeouts and cooperation [9]. These inform identity pinning, replay protection, and state-machine reasoning. They do not justify copying a timeout refund into a destination that can still accept a delayed export. A future Rldcoin re-export must first define asset lineage, destination finality, local debit before onward credit, and non-duplication through cyclic routes.

Offline and asynchronous payment alternatives. Teechain reduces dependence on timely blockchain access using trusted execution and replication assumptions [16]. Hu et al. study intermittent rural connectivity with bank-supported local payments [17]. The BIS Polaris handbook surveys offline CBDC design decisions [22]. These are useful alternatives to compare, but neither a trusted device nor a bank-backed claim is an assumption-free offline bearer coin. Copyable software state and a signature alone cannot stop two isolated recipients accepting conflicting spends. Any future offline mode must specify preallocated spending rights or hardware/trust assumptions, value and duration limits, reconnect behavior, and who bears an unreconciled loss.

Interplanetary proposals. Puente and Puente's 2025 preprint proposes Bitcoin-related transit receipts and delay-aware operating policies [18]. Its auditable carriage idea is relevant; its preprint status and planetary scope limit what it establishes. Rldcoin can investigate signed courier receipts as operational evidence without making their timestamps a consensus clock or importing claims of automatic BPv7 custody. Transport claims must be checked against the RFCs themselves. No cited preprint validates Rldcoin, establishes interstellar deployment, or removes the light-time lower bound.

15. Route capacity and conformance evidence

Capacity before availability claims. A route profile must publish the intended contact plan, payload and archive sizes, admitted arrival load, per-hop bandwidth, storage quotas, retry policy, and forward/return reachability. Capacity-aware routing research explicitly considers contact and buffer constraints [19]. For an illustrative fluid queue with admitted serialized rate r bytes/second, a no-service gap D , and burst allowance B , at least $r \cdot D + B$ bytes are needed at that bottleneck before replication, metadata, and safety margin. Recovery service must exceed ongoing arrivals long enough to drain the backlog. Average bandwidth alone does not guarantee a particular delivery deadline or survive a failed contact.

The reference 256 MiB queue would hold only about 2,684 seconds, or 44.7 minutes, at an assumed 100,000 serialized bytes/second with no departures. This is dimensional arithmetic, not a measured traffic rate or throughput result; frame-count limits and overhead can bind earlier. A four-year outage cannot be supported by citing a bounded terrestrial queue. Admission must distinguish stored bytes from promised future capacity, reject or backpressure excess traffic, and keep ledger exports locked even if every relay reaches its retention limit. Retrying and batching history pages may reduce overhead; they cannot waive missing ancestry.

Reproducible experiments. Each result should identify genesis and rule hashes, code revision, topology, independent owners, contact trace, random seeds, hardware, initial state, offered load, and injected faults. Scenarios should include short contacts, asymmetric links, the NASA-inspired multi-week gaps, multi-year logical-time gaps, and permanent separation. Accelerated simulation measures protocol behavior under a model; it cannot validate physical links, years of media aging, or independent custody. Report destination spend latency separately from the latency until the source learns

it. Publish success fraction, unresolved exports and their age, duplicate-credit violations, queue high-water mark, bytes per completed import, CPU/replay cost, and recovery outcomes. Latency percentiles over successful transfers alone must not hide undelivered transfers.

Acceptance conditions. Duplicate and reordered delivery must never produce extra credit. Lost receipts, expired bundles, exhausted queues, and permanent partitions must never cause a refund. A source checkpoint conflict must preserve both certificates and stop affected new value transitions once detected; the report must retain any already-paid exposure. Destination reorganization must invalidate receipts tied to orphaned history and reconstruct balances consistently. Missing signers must stop new source finality without inventing signatures. Restart, disk corruption, key/lock restoration, wrong destination, and unknown rule versions must have explicit rejection and recovery evidence. Conservation should be checked after every simulated transition and every recovered state. A finite test campaign can find counterexamples; it is not a proof over all executions.

16. Mandatory interstellar payment requirements

These requirements define conformance to the interstellar payment design. They are mandatory alongside the A-G foundational acceptance phases and N1-N10 relay obligations embedded in Section 20. They apply to an exact adopted version; listing them does not establish implementation or operating qualification. Every requirement needs a versioned specification, stated fault assumptions, actual node evidence and independent review.

I1. Currency and regional identity. All regions must share one authenticated currency root while retaining distinct regional genesis and rules. A locally trusted root and carried adoption evidence must suffice to validate regional admission, rule commitments and signer epochs without an online Earth directory. A discovered identity, self-signed region label or route advertisement must confer no ledger, finality or issuance authority. Unknown currency roots, genesis identities, rules and unauthorized regions must be rejected.

I2. Local autonomy during remote disconnection. With sufficient local communication and security resources, a region must start, progress, receive and spend locally verified value, and recover consistently while every remote-source endpoint is unavailable. Exports already covered by locally verified finality may be imported; missing new evidence blocks only the transitions that need it. Remote freshness must not gate all local balances or payments. New contact evidence must be authenticated, replayed and installed without erasing prior finality or hiding conflicts. Full CLI startup, actual signed payments, block production and persistent recovery are required; a fixed-cache interface test is insufficient.

I3. Unique issuance and conservation. The signed currency root must bind its cap, integer units and issuance authority. The currency specifies 100 billion RLD, 10^{24} runlai per RLD and zero initial allocation, bound by exact signed adoption. Only the authorized origin issues native currency; additional regions have zero native issuance. No new reserve is created by admission or import. For compatible selected histories, $I = U + E + T$ must hold through splitting, merging, change, fees, channels, import, onward export and recovery. Historical proofs, receipts and import tombstones are evidence, not additional balances. Delegating issuance would require a separate non-overlapping budget protocol and is not a dependency of this design.

I4. Transfer between arbitrary authorized regions. Every qualified region must be capable of both export and import, including multiple authorized source regions. An owner must authorize local debit before a finalized export can create unique credit at its exact destination. An export must bind its currency root, source and destination genesis, rule and finality epoch, input dependencies, amount, recipient, fee/change rules and unique domain-separated ID. Full authenticated history or an explicitly adopted equivalent must establish all dependencies. Wrong domains, missing evidence, altered recipients, duplication and reordered carriage cannot create value.

I5. Onward export and composed finality. Imported value must be usable for local payments and for a new export. The exporting region's recognized finality must protect the imported input, its accepted provenance and the new local debit before onward import is allowed. Six-block probabilistic maturity is not irreversible export finality. Splits and merges require a finite well-founded causal dependency graph rooted in authenticated issuance; self-reference and cycles are rejected even when the geographical route returns to its origin; repeating historical ancestors must not issue value.

Reorganization before finality, conflicting ancestry and recovery must preserve the same conservation rule. This requires a general regional ledger and a validated finality composition.

16. Actual return of value. A return is a new owner-authorized export from the region holding the asset and a new unique import into the earlier region. Each leg retains its own ID and permanent duplicate-import protection. Returning information, a custody receipt or the asset itself must never unlock the first source debit again. A three-ledger onward-and-return path is required for conformance testing; Earth-to-Proxima-to-Andromeda-to-Earth supplies illustrative labels without asserting a physical route. Lost receipts, expired transport copies and permanent partitions cannot refund an export.

17. Regional finality and trust evolution. Finality must complete within a region under a specified local fault model, with no simultaneous vote across stars. The adopted protocol must define signer independence, quorum intersection, durable locks, view changes, epoch transitions, reconfiguration and conflict recovery. A threshold alone is insufficient: established BFT protocols also specify voting and locking rules [14]. Four keys under one owner are not independent Byzantine tolerance, and changing four-of-four to three-of-four is not a qualified protocol. Detected authenticated conflicts must preserve both histories, quarantine affected existing local balances and every provenance descendant, including split/merge/change/fee/channel outputs, and suspend all their new local spends, imports and onward exports under Section 18/S8-S9, and propagate evidence. A disconnected region may not yet know a conflict; already-paid exposure cannot be promised reversible. Reconfiguration and custody need independent adversarial tests and explicit adoption.

18. Native discovery and causal evidence carriage. Every normally started full network node must enable authenticated discovery and bounded durable relay in the same lifecycle, without a separate relay launch or mandatory central Earth service. With usable contacts it must progressively learn neighbors and candidate distant routes, automatically forward admitted payment evidence for other participants, and resume retained work after restart. Qualification must demonstrate incremental node addition extending reach, a usable alternative path delivering after relay loss, and capacity refusal preserving already admitted evidence. Additional node count alone must not be reported as increased consensus security or guaranteed delivery. Section 20's N1-N10 apply to every adopted node's integrated discovery and relay lifecycle. Real contact adapters, authenticated retention/deletion obligations, resource admission and recovery must be qualified. Transport must respect physical contact and light time; forward and return reachability are separate. Bad neighbors, asymmetric links, restarts, lost copies and full queues must not change ledger value. A mesh receipt is neither import acceptance nor spendability.

19. Recipient-verifiable payment states. Wallets and nodes must separately expose discovered identity, candidate route, observed contact, transport receipt, ledger import, local maturity, export finality and source knowledge of a returned receipt. The recipient must verify the currency, region, ownership and accepted local state independently. Locally spendable imports must not wait for Earth or a returned acknowledgment. Displayed stale information cannot substitute for current signing qualification. Unknown dependencies and detected conflicts must remain visible rather than being represented as settled money.

110. Durable history and bounded recovery. Each region must qualify its active-history and archive design, authenticated state/proof recovery, permanent unique-import commitments, finality locks and signer anti-rollback protection. Checkpoints alone cannot waive history under rules that still require replay. Multiple independently held archives, corruption and old-backup tests, resource limits and measurable restoration cost are required beyond the first 200,000-block issuance era and the declared operating scale. A lost tombstone or old snapshot must never make a previously imported export importable again. The qualification states a storage and preservation horizon, not infinite media life.

111. Cryptographic longevity and delayed revocation. Suite/parameter identifiers, canonical signature/hash formats, purpose domains, key roles/epochs and finite verification horizons must be authenticated and versioned. Independently reviewed post-quantum authorization and transport migration, downgrade rejection, overlap/retirement rules, still-trustworthy evidence renewal, recovered trust anchors and in-flight-export interpretation are mandatory obligations [11, 21, 26-29]. Tests must cover compromise, late revocation, stale backups, lost keys/archives and competing upgrade histories while preserving conservation, finality, caller locks and consumed-export IDs. Distant compromise cannot

become known faster than communication; renewal cannot rehabilitate lost authenticity. Out-of-scope evidence remains preserved but unavailable for new acceptance until an authorized verification rule resolves it. A suite outside its adopted verification horizon cannot satisfy this obligation. The 100-million-year continuity objective and Andromeda topology establish no cryptographic lifetime guarantee.

I12. Independent qualification and sustained service. Independent operators, separate finality custody, external security review and reproducible recovery must substantiate the exact release. Default relay service must declare its storage, bandwidth, admission, retention and operating-cost budgets; starting a node earns no automatic reward or issuance right. Local mining/finality, long-haul carriage and archive preservation need explicit resource and fee budgets; zero-native-issuance destinations do not inherit the source mining budget. User payments, unilateral local-channel exit, device recovery and anti-rollback must be verified under the declared scale. Reports must retain unresolved exports, incidents, paid exposure and budget shortfalls. Every physical route and preservation period must have its own evidence; operator telemetry and accelerated logical-time tests cannot establish independent custody or physical interstellar availability.

17. Protocol composition and assumptions

A regional adoption package must bind the currency root, regional genesis, precise rules and implementation, admission authority, local finality protocol/epochs and upgrade ancestry. Directory answers are candidate data. A verifier authorizes them from its adopted local trust context and carried evidence, without a mandatory online central lookup. Evidence updates must be checked before becoming trusted state, while conflicting evidence is retained and isolated according to declared dependencies.

The target value path is:

```
locally spendable output
-> owner signs export -> local debit -> recognized local finality
-> asynchronous evidence carriage -> exact destination validation
-> unique import -> local maturity -> local payment or new export
-> onward region; return is another finalized export and import
```

Composition assumptions. Regional ordering and authenticated asynchronous export evidence support causal transfer when finality is sound, complete provenance remains available, local participants can progress and a usable contact path eventually carries evidence within its qualified verification horizon. The adopted profile must provide a composition proof for arbitrary multi-source dependency graphs and cyclic returns, independently operated governance and anti-rollback custody, persistent history beyond its declared issuance-era and operating scale, qualified cryptographic renewal, bounded resource funding and an adapter appropriate to the route. Queue and history limits must cover the admitted service; each cryptographic suite must cover the finite accepted horizon. No regional quorum protocol or faster local channel removes physical propagation delay.

Incompatible promises and explicit choices. There is no guaranteed finite remote delivery when contact may never resume, no immediate globally current balance during disconnection, and no automatic safe refund from silence. A local partition without sufficient ordering resources may stop payment finality. Global conflict/revocation awareness cannot precede arriving evidence, and an expired authentication horizon may block acceptance of preserved value. These choices retain I1-I12 as obligations for qualified routes and periods without promising infinite availability, universal trust-free admission or million-year security. Liquidity advances from prepositioned local funds remain separately disclosed credit.

The export certificate must cover accepted input dependencies and the debit, not merely a block number or operator statement. Ordinary local spending remains subject to the region's disclosed reorganization rules. Export safety additionally requires the adopted local finality protocol to constrain selected history, restart, subsequent certificates and epoch changes. The profile must supply a composition proof, complete canonical wire formats and adversarial conformance evidence; a diagram or threshold configuration cannot supply them.

No timeout refund. Cancellation is optional and disabled by default. A future cancellation would require recognized destination finality that permanently consumes the export ID and excludes both existing and later import, followed by

source verification and explicit adoption. A query, old non-membership proof, transport expiry or absence of a receipt is insufficient. Local liquidity advances, proof compression and fully offline bearer spending may be separate proposals; they cannot replace I1-I12 or remove their trust assumptions.

Mandatory end-to-end campaign. At least three actual native regional ledgers and multiple owners must execute split/merge/fee transactions, multiple-source imports, onward exports and a cyclic return. Only neighboring contacts are provided. After origin authorization/export, all Earth services and directories are removed: distant local payments and onward legs must continue; the return-to-Earth leg may wait for restored contact. Earth then imports that new export exactly once without releasing the initial debit. Full CLI startup, actual local block production under the adopted consensus (proof-of-work mining where applicable), maturity and restart must work without source HTTP. Tests must cover pre-finality reorganization, post-maturity onward export without finality, deep locked forks, signer faults, equivocation and dependency quarantine, in-flight epoch rotation, delayed revocation, wrong destinations, malicious neighbors, queue exhaustion, lost acknowledgments, old backups, lost archives and permanent partitions. Conservation is checked at every transition and restoration.

Cryptographic acceptance evidence. I11 requires an exact adopted suite and code version, independent cryptographic/implementation review, published known-answer and malformed-encoding vectors, purpose and network replay rejection, hybrid component failures and downgrade rejection, and measured full transaction/quorum/archive sizes and verifier cost. Multi-region campaigns must carry exports across overlapping and retired epochs, simulate delayed revocation and conflicting governance, recover stale backups without resetting locks, renew still-authentic history before expiry, reject renewal of already-untrusted evidence and preserve conservation through key loss and archive loss. Use logical-time tests to check policy behavior; they cannot prove that an algorithm or medium survives actual centuries. Qualification states the finite horizon and requalification triggers, never a 100-million-year security certificate. A finite local experiment cannot substitute for this evidence.

Qualification scope. Protocol conformance requires an exact authenticated profile to satisfy A-G and I1-I12 under published fault assumptions, capacity, verification horizons and independent evidence. Monetary adoption additionally requires a signed zero-initial-issuance genesis and a verified release; test balances and keys confer no monetary assets or authority. Physical-route qualification applies only to a specific measured route, its operators, contact/capacity trace and operating horizon. A protocol proof or local experiment cannot establish that a stellar service is operating. Each claim must remain within its independently supported scope.

Regional block intervals, maturity and channel challenges must be selected from local propagation, persistence and attack measurements and authenticated together. Stellar light time or an accelerated test clock cannot choose them. A changed route, scale, trust model or verification horizon requires renewed qualification. Conserved value, local autonomy and arbitrary authorized-region transfer remain conformance requirements under conditional delivery and explicit failure outcomes.

18. Normative architecture and transition rules

S1. Authority and interpretation. This paper specifies the required protocol. Mandatory means that a release cannot claim compliance without the stated behavior and evidence. Sections 5-9 describe a proof-of-work and unanimous-checkpoint regional reference profile; their proof-of-work, four-of-four signatures and block delays illustrate a profile rather than constrain every future implementation. Sections 16-21 define the target contract. A target rule overrides a narrower reference example for a new adoption; it never retroactively alters an already signed genesis. A deployment needs a separately authenticated, complete executable protocol profile specifying encodings, consensus messages, cryptographic suites, limits and activation. Producing that profile and proving its composition are release gates, not discretionary research. An implementation detail, implementation limitation or mutable external plan cannot lower this contract.

S2. Trust and admission. CurrencyRoot binds zero allocation, the 10^{35} -runlai cap, origin genesis, issuance recurrence, allowed regional-admission authority, governance constitution and initial suite policies. RegionAdmission binds CurrencyRoot, a unique region genesis, complete rule/profile digest, zero regional issuance, validator membership and finality policy, accepted source dependencies, qualification horizon and governance succession. A verifier pins its initial

root through independently checked adoption; root discovery is not trust establishment. Admission authorities must have separate independently held keys and an explicitly adopted approval rule. Every deployed approval/recovery role must have authenticated identities, quorum, scope, predecessor, conflict/activation rules and a fail-closed default in the signed adoption package before qualification; operators cannot fill an omitted authority during an incident. Public applications, reproducible validation, published controller/conflict disclosures and authenticated decisions make admission auditable. Naming a region, forwarding its advertisement, owning coins or creating validator keys confers no voting or issuance right. Permission to discover and relay remains distinct from permission to finalize. Admission is a disclosed trust boundary, not an assumption-free permissionless Sybil solution [37].

S3. Regional consensus. The target baseline is independently operated Byzantine regional finality with a fixed membership epoch: $n = 3f + 1$ active equal-weight validators, at most f Byzantine, and at least $2f + 1$ distinct approved votes in each required quorum. Larger or weighted sets require their own intersection and fault-bound proof. A complete adopted locking/view-change protocol must be reviewed; a signature counter alone is invalid. Proposals bind currency, region, epoch, height, view, parent, rules, state root and complete command commitments. A signer must validate the proposed state, persist the protocol's lock and complete signing intent before releasing a vote, refuse conflicting or regressive requests and retain the lock across restart and certified view changes. Finality is installed atomically with the exact certified ledger extension. Missing quorum stops finalization; a transport receipt or time passage never substitutes for votes. Safety is conditional on fault bounds, honest locks and authentic keys. Progress additionally requires the adopted protocol's eventual local synchrony and available honest quorum; remote contact is unnecessary for unrelated local payments [12, 14]. The proof-of-work reference remains separately labeled probabilistic before its checkpoint rule.

S4. Epoch transitions. An epoch change closes a precisely committed prefix and binds old/new membership, rule hashes, suites, locks and activation. The old epoch authorizes the closing transition under its adopted quorum; the new epoch acknowledges the identical closed prefix and carried locks under its own quorum before it can finalize new value. The protocol profile must prove that no old continuation and new continuation can both be accepted as conflicting finals, including offline validators and delayed votes. Continuing validators retain their full prior restrictions; new validators establish an independently witnessed current head. No validator may run copied signing custody in parallel. Unavailable closing authority makes reconfiguration unavailable. A new signer set, threshold reduction or reset genesis cannot recover existing funds by decree. HotStuff is a design reference, not evidence that this reconfiguration composition is already proved [14].

S5. Value objects and arithmetic. Every output has an immutable currency/region identity, command identifier and output index, positive integer amount, owner/recovery policy, maturity rule and authenticated provenance commitment. Amounts are unsigned integers bounded by the cap; parsing rejects fractions, signs, noncanonical digits, duplicate keys and out-of-range values before arithmetic. All sums, products, fees and issuance use checked arithmetic with a specified width of at least 117 unsigned bits for the cap, and additional checked width for intermediate operations; using a 128-bit type alone does not excuse unchecked accumulation. A local transition consumes each distinct available input once and requires $\text{inputs} = \text{outputs} + \text{explicit fees} + \text{explicit channel/export allocations}$. Every fee has a named beneficiary or escrow; no implicit destruction/recreation or negative change is allowed. Mint is accepted only on the authorized origin under Section 6. A fee transfer is not a second mint. Speculative/immature or quarantined inputs cannot fund an accepted spend.

S6. Export and import. An ExportIntent binds suite/domain, currency root, exact source/destination genesis, rules/epochs, selected input IDs, recipient policy, gross amount, net amount, fees, change and nonce. ExportID is a domain-separated commitment to that canonical intent and its selected source position; rewriting a destination creates another intent requiring another legitimate debit. The source atomically consumes available inputs and records the pending export plus fees/change. Only a finality certificate covering that debit and all its input dependencies makes it exportable. The destination reconstructs and validates the complete admitted source history and finality closure, checks its own exact identity and recipient, and atomically records ExportID consumed together with the uniquely credited output. Concurrent duplicates, restart and delayed copies return the same accepted result or refuse; they cannot add credit. Import becomes spendable under the destination's authenticated finality/maturity profile. A return receipt binds the exact import history but neither credits the source nor unlocks its debit. After import, a new onward export follows the same

local consume/finalize procedure. Cancellation is excluded from the baseline; Section 12 describes obligations for any separately adopted future cancellation.

S7. Well-founded provenance. The evidence dependency graph is a finite directed acyclic graph rooted in authenticated origin issuance and admitted histories. Local parent references strictly advance the region's ledger order; a command may depend only on already validated predecessors. An import depends on an earlier finalized source debit, and an onward debit depends on its earlier accepted local import and subsequent transitions. The proof verifier must reject self-reference, ancestor cycles, missing roots, duplicate consumption and inconsistent histories, with explicit object, decompression, graph-depth/fanout, duplicate-ancestor, signature-work, CPU/RAM/disk and recovery-index limits and a pending state when authentic closure is incomplete. A route Earth to Proxima to Andromeda to Earth can be cyclic geographically while each successive value transition has a new ID and acyclic causal predecessors. A ring of imports cannot authenticate itself. No remote timestamp or globally synchronized clock is needed to topologically verify complete carried evidence. Proof compression cannot waive these checks without a separately adopted and reviewed validity construction.

S8. Conservation and provenance through mixing. Across compatible selected regional prefixes, $I = U + E + T$ and $I \leq 10^{35}$ runlai. The buckets are mutually exclusive: U includes immature and quarantined unspent outputs but excludes every amount already counted in E or T; E includes live channel escrows and explicitly adopted fee reserves; T counts every export debit present in the selected source history, including pre-finality pending exports, until its unique destination credit, with gross/net fees assigned once. Only a finalized debit is eligible for import under S6. If an unfinalized debit is orphaned, replay atomically reverses that transition and restores its source inputs; a finalized debit cannot be undone by ordinary reorganization. Amounts unavailable to users remain liabilities, not deleted supply. Every output produced by payment, split, merge, change, fees, channel settlement, import or export carries a commitment to the union of its input provenance. The baseline deliberately taints the entire resulting output if any input lineage becomes disputed; proportional segregation is an optional future profile requiring a new conservation and isolation proof. Nodes must retain an authenticated reverse-dependency index or reconstruct it from complete history. Indices accelerate incident discovery and never establish validity alone. This conservative mixing rule limits spendability during incidents but prevents contaminated change or fees from escaping isolation.

S9. Conflicts and resumption. A valid contradictory finality or governance pair is recorded as an incident with both statements, identities, conflict scope and adopted policy. On detection, the verifier marks all dependent outputs, escrows, pending imports and onward exports quarantined, including previously received local balances and every reachable descendant through split/merge/payment/fee transitions. No quarantined input may make a new local spend, channel update, close creating available value, import credit or export; unrelated proven lineage may progress under its own regional quorum. The verifier publishes a sanitized exposure record and propagates authenticated incident evidence with priority. Detection cannot reverse payments made before notice and cannot immediately inform isolated regions. Resumption requires an authenticated resolution under the pre-adopted incident authority, preserving spent IDs, custody locks, liabilities and a replayable state transition. Losing records or replaying an earlier backup must not clear quarantine. Resolution cannot introduce hidden issuance or erase a contradictory certificate. If authority or a conservation-preserving resolution is absent, affected value remains unavailable indefinitely.

S10. Governance under disconnection. The adopted constitution must separate admission, consensus membership, software/profile upgrades, cryptographic renewal, owner recovery and incident resolution into explicit approval roles with independently held custody, succession and conflict rules. Changes name their predecessor, scope, evidence, exact successor policy and activation conditions. Local ratification is an authenticated ledger transition, never automatic acceptance of the latest download or largest advertised epoch. An isolated region continues only within its already qualified local rules and trust horizon; it cannot discover an unseen revocation. New imports require a compatible authenticated policy chain. After reconnection, compatible histories may exchange evidence and continue; incompatible finalized ownership or governance branches cannot be merged by set union, longest-chain preference outside the profile or deleting spent records. They remain distinct disclosed adoption branches with dependent value quarantined until authorized resolution. Owners may voluntarily migrate legitimately controlled unencumbered value under an adopted rule, but an operator cannot force a synthetic global reunification. Offline governance therefore chooses bounded authority and conditional interoperability over universally simultaneous upgrades.

S11. Freshness and recovery. A signed old backup proves a prior authentic state, not that no later spend, lock, key rotation or revocation exists. Restoring value-serving or signing custody requires a surviving independently maintained monotonic witness of the last acknowledged signer/caller head, finalized ledger prefix, consumed-export commitment, governance/suite floor and incident state. Its complete authenticity, independence, durability and anti-rollback mechanism must be qualified; a hash stored alongside the same rollbackable backup is insufficient. The restored state must prove a compatible extension to that witnessed head before issuing signatures or reporting available value. Lack of an adequate witness keeps the recovery read-only and unavailable for spending; obtaining a newer website timestamp or resetting a private directory does not repair it. Multiple independently administered witnesses or qualified monotonic hardware can reduce common-mode loss, but neither can be assumed eternal. Witness loss, copied-key concurrent custody, interrupted restore, stale channel state and rollback of all local files are required adversarial cases. Recovery must never first-sign from a retained response or import twice.

S12. Owner succession. Owners may pre-adopt a purpose-separated recovery/inheritance policy while their current authority remains trusted: explicit guardian identities, threshold, replacement scope, revocation rules, local challenge conditions and how balances/channels move. Recovery authorizations bind the old policy, exact new key and ledger position and consume a unique recovery nonce. Guardian claims, civil-law documents and proof of death are not native consensus facts unless an adopted oracle/trust policy explicitly makes them so. A challenge delay helps only under its stated local observation and inclusion assumptions; no finite delay protects an indefinitely unreachable owner. Users can select direct-key custody with unrecoverable loss or a reviewed guardian policy with its disclosed collusion/availability risk. Unknown heirs, broken old keys or unsigned operators receive no automatic ownership. The protocol does not promise anonymity, legal enforceability, universal inheritance or loss-free recovery.

S13. Custody and retention. Relay admission declares payload identity, byte/work budget, retention obligation, recipient/route restrictions and expiry semantics under a locally qualified clock or event policy. Custody is acknowledged only after full transport authentication and durable bytes/metadata; it is not proof of valid money or end-to-end delivery. A relay refuses new work before exhausting protected resources and applies fair scheduling/priority without evicting obligations it already accepted. A courier copy may be reclaimed only according to its disclosed contract, after durable authenticated handoff or the agreed retention condition; an expired carrier receipt reports expired custody, never settlement or non-import. Source owners and independently funded archives retain the canonical export and required history for the monetary/dispute horizon, regardless of courier expiry, and permanent consumed-ID obligations survive compaction. If no archive can cover an intended journey's horizon, the route cannot be qualified for new value. Permanent contact loss can still strand an export. This reconciles bounded carrier storage with indefinite monetary pending state; neither entails an unlimited free relay promise.

S14. Resource and service budgets. Every adopted region/route declares byte/count/depth/verification/connection limits, admitted load, queue service model, contact trace, backlog margin, restart cost and explicit overload behavior. Object/decompression limits and deduplicated ancestor traversal bound parse and graph work; a proof profile additionally bounds fanout, signature count, CPU/RAM/disk and recovery-index cost. Cheap parse/length/purpose checks precede expensive signature/replay work; per-peer quotas, bounded authenticated scheduling, multiple independent contact paths and locally accountable admission resist denial of service without discarding accepted evidence. Sybil relay identities never gain ledger votes. Origin subsidies do not fund destination consensus automatically. Regional fees, voluntary operator funding and separately agreed carriage/archive contracts pay their respective services; a prepaid service claim is accounted escrow, not newly issued RLD. Budgets include cryptographic renewal and media refresh. If funding/capacity/fault independence falls below the qualified policy, the service stops accepting new obligations and publishes the deficit. No fixed cap or fee market proves that people will operate a region for a hundred million years.

S15. Privacy and confidentiality. The baseline is a publicly verifiable pseudonymous ledger, with disclosed amounts, provenance and potentially identifying route/timing metadata. Wallets minimize address reuse and logs; relays disclose only necessary carriage data and use independently reviewed authenticated encryption where contact permits. End-to-end payload encryption must still allow designated validators to obtain required verification bytes and must not conceal incident dependencies from them. An encrypted packet is not valid value. PQ key establishment addresses future decryption risks only within the adopted suite's horizon; it cannot erase information already disclosed. Anonymity sets,

traffic padding and confidential amounts are separate costly proposals requiring resource measurement and conservation/provenance proofs before adoption. No user-facing anonymity claim is permitted for the baseline.

S16. Hardware and archive continuity. Independent regions, power domains, media/implementations and geographic custodians reduce correlated failure. Archives preserve complete original bytes, decoding/validation tools, trust/epoch ancestry, finality/consumed-ID records and authenticated renewal history. Custodians periodically verify complete recoverability, refresh media and perform destructive-loss drills on expendable replicas. Radiation damage, lost clocks, entropy failures, device theft, unavailable spares, malware and total institutional disappearance belong to the qualification fault model. No particular storage medium, hardware enclave or duplicated cloud account establishes generational independence. Complete loss of authenticated history and surviving trust/freshness witnesses is a declared unrecoverable state; fabrication, old-key re-signing after compromise and a new genesis cannot recreate ownership. Operational continuity is a maintained human system rather than a finite experiment proving future civilization.

S17. Software, encoding and release security. A protocol profile must define canonical bytes, integer bounds, domain separation, error states and deterministic replay independently of one language implementation. At least two independent verifier implementations and cross-platform vectors must agree on all accepted/rejected transitions. Parsing, signature validation, state persistence, cryptographic randomness, wallet intent and relay exposure need fuzzing and independent adversarial review. Reproducible builds, pinned dependency/source commitments, signed provenance, least-privilege execution, rollback floors and maintained vulnerability/incident records constrain supply-chain substitution; a signature on an executable is not proof that it is safe [38]. Builds and experiments have isolated no-value state. Software installation does not change adopted monetary rules. Emergency patches follow authenticated scope/activation and retain evidence; unsafe value operations stop while review proceeds.

S18. Finality shown to users. The wallet distinguishes constructed, partially authorized, fully authorized, queued, locally included, source-finalized/export-committed, transport-custodied, destination-evidence-verified, imported-immature, locally spendable, onward-export-eligible, quarantined and unqualified/unrecoverable states. Each display names actual currency/region, amount/net fee, evidence/head, maturity/finality rule, known incidents and knowledge timestamp. Source knowledge and recipient state are separately represented. A time estimate uses an explicit contact/service model and includes unresolved probability; it is never a settlement promise. Signing revalidates current local qualified state and ownership rather than trusting a stale display. Credit advances and hardware-assisted offline receipts must be labeled separate claims with counterparties, limits and loss allocation. The baseline has no unrestricted offline bearer spending between mutually isolated recipients. Full region partitions stop quorum-dependent operations; a healthy local region may still spend proven unrelated value when distant regions are absent.

19. Risk register and falsifiable controls

This register is part of the mandatory contract. Every record names a threat, an assumption, a required mechanism, a residual risk and an acceptance test; its mechanism and acceptance conditions are mandatory for the corresponding qualified profile. Passing a finite test does not remove a residual risk. The S, I, N and A-G references are defined in this paper. Each qualification report must attach its assumptions, thresholds, failure evidence and requalification triggers to the corresponding record; unresolved records cannot silently become accepted.

R1. Light time, missing contact and asynchronous termination. Threat: evidence never arrives or cannot arrive before a promised deadline. Assumption: safety needs authentic records, while liveness needs actual eventual contact and regional progress. Mechanism: S6/S13 distinguish monetary pending state from delivery, preserve exports and expose uncertainty; no timeout refund. Residual: permanent stranding and unbounded latency remain possible [2, 10, 12]. Acceptance: inject permanent/asymmetric separation, lost acknowledgments and logical multi-year gaps; no extra credit, refund or fabricated remote state, and reports retain unresolved exports and their age. Physical-route claims require A-G/G and N9/N10 evidence for that exact route.

R1 operational obligations. Detection/check: missing contacts and unresolved-export ages. Affected value/scope: pending exports. Recovery authority: adopted source/destination authorities; no refund authority; exact identities, quorum and scopes must be signed before qualification. Stop rule: dependent delivery promises and out-of-horizon new exports.

Resource payer: sender and contracted carriers/archives; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R2. Local partition and Byzantine consensus. Threat: conflicting finals or starvation. Assumption: S3's approved independent membership, at most f Byzantine, authentic keys and durable locks; liveness additionally needs the reviewed local timing condition. Mechanism: validated locking/view-change protocol, intersecting quorums and halt without quorum. Residual: excessive compromise, correlated custody or indefinite local partition can defeat safety assumptions or stop service. Acceptance: systematic equivocation, delayed/reordered votes, leader loss, split quorums, restart and network recovery against exact profiles; no two conflicting finals under the claimed bound, and bounded progress only inside the stated synchrony/load model. Model/implementation correspondence and independent review are required, not just a three-signature demo [14].

R2 operational obligations. Detection/check: authenticated conflicting votes/finals or quorum/timing loss. Affected value/scope: region and dependent lineage. Recovery authority: signed epoch/incident authority; exact identities, quorum and scopes must be signed before qualification. Stop rule: new finality and its dependent value operations. Resource payer: regional fee beneficiaries and disclosed sponsors; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R3. Sybil identities and concentrated control. Threat: one controller masquerades as independent validators, couriers or archives. Assumption: admission/custody disclosures and identity verification are trustworthy within a stated institutional model. Mechanism: S2 separates open relay discovery from authenticated voting membership, with public admission, conflict disclosures and independently administered keys/resources [37]. Residual: hidden collusion, coercion and authority capture remain. Acceptance: unauthorized self-admissions and duplicated identities gain no votes; independent parties reproduce key ceremonies and failures; reports disclose actual common controllers and funding dependencies. Stake/PoW weighting is not silently substituted for this admission model.

R3 operational obligations. Detection/check: admission violations and controller/custody audits. Affected value/scope: controlled region and its dependent claims. Recovery authority: pre-adopted admission/incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: unqualified membership acceptance. Resource payer: admission participants and regional operators; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R4. Cross-region duplication and counterfeit provenance. Threat: duplicate import, wrong destination, self-certified rings or double consumption through cyclic routes. Assumption: authentic finality and complete well-founded histories. Mechanism: S6/S7's atomic debit/credit, immutable IDs, destination binding, consumed-ID permanence and acyclic rooted dependency closure. Residual: key/finality compromise and missing closure prevent trusted acceptance. Acceptance: arbitrary multi-source split/merge/onward/return workloads, concurrent duplicates, cross-currency replay and self/ancestor cycles; conservation at every transition, zero duplicate credit and explicit pending/refusal for incomplete closure. A formal composition argument plus real native campaigns is mandatory.

R4 operational obligations. Detection/check: invalid closure, duplicate tombstones and failed conservation. Affected value/scope: candidate import and dependent lineage. Recovery authority: adopted region/incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: candidate credit/export with incomplete or invalid proof. Resource payer: sender and verifying region under declared proof fees/budgets; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R5. Detected conflict spreading through local value. Threat: contaminated balances escape via payments, change, fees or channels after detection. Assumption: complete authentic provenance and incident index/records survive. Mechanism: S8/S9 quarantine every dependent descendant and forbid its next value transition; unrelated lineage continues. Residual: already paid losses and disconnected recipients unaware of the incident cannot be instantly repaired. Acceptance: inject conflicts after mixed-input local payments, split/merge, channel funding/closing and onward export; every reachable dependent output becomes unavailable, survives restart/old-backup attempts and cannot fund new local transfers. Both certificates and exposure remain public; no inflationary compensation.

R5 operational obligations. Detection/check: authenticated contradictory statements and reverse-dependency traversal. Affected value/scope: all reachable descendants including mixed local outputs. Recovery authority: pre-adopted incident authority; exact identities, quorum and scopes must be signed before qualification. Stop rule: every new value transition consuming disputed lineage. Resource payer: affected regional operators and funded incident archives; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R6. Reconfiguration and disconnected governance. Threat: old/new signer sets or valid competing constitutions authorize conflicting histories. Assumption: surviving pre-adopted approval/succession authority and complete transition ancestry. Mechanism: S4/S10 old/new closed-prefix approvals, lock carryover, local ratification and incompatible-branch quarantine. Residual: missing quorum or irreconcilable adoption can produce permanent unavailability or distinct communities. Acceptance: delayed closing votes, simultaneous rotations, interrupted activation, lost authorities, conflicting branch reconnection and stale floors; no conflicting continuation, automatic set-union merge, lock reset or online-Earth dependency. Any proposed resolution must replay liabilities under the adopted policy.

R6 operational obligations. Detection/check: conflicting/incomplete epoch or governance chains. Affected value/scope: branch-dependent balances and new acceptance. Recovery authority: signed closing/new epoch and governance/incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: conflicting or unwitnessed transitions. Resource payer: adopting regions and governance participants; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R7. Quantum or classical cryptanalytic failure. Threat: forged ownership/finality/governance or exposed recorded traffic. Assumption: each exact suite remains trustworthy for a finite accepted horizon. Mechanism: Section 11/I11 authenticated suite agility, independently reviewed PQ signatures/KEMs, no downgrade, pre-compromise migration and renewed archives. Residual: no known suite has an eternal guarantee; prior disclosure and already-lost authenticity cannot be repaired. Acceptance: standard vectors, malformed encodings, component failure/stripping, parameter/domain confusion, old-suite retirement and real full-proof bandwidth/verification measurements. Simulated break/expiry gates stop new dependent acceptance rather than accept an old forged-capable key. Test results establish policy behavior, never the date or absence of future attacks.

R7 operational obligations. Detection/check: published review alerts, expiry and invalid suite/parameter evidence. Affected value/scope: keys, balances and histories protected by that suite. Recovery authority: pre-adopted suite/renewal and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: new authorization/acceptance lacking qualified authenticity. Resource payer: owners, regions and renewal/archive contracts; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R8. Delayed revocation and stale trust. Threat: a distant verifier accepts a compromised key before notice. Assumption: qualified trust/cryptographic horizons and authentic update chains, with no instant global revocation. Mechanism: S10's scoped revocation/activation and Section 12's carried renewal ancestry; stale/unsupported new evidence stays pending. Residual: bounded-horizon acceptance still has exposure to compromise whose notice has not arrived. Acceptance: deliver revocation before, during and after imports across overlapping/retired epochs and permanent separation; preserve history, consumption and paid exposure, enforce declared scope and never represent silence as current non-revocation.

R8 operational obligations. Detection/check: authenticated scoped notice or horizon expiry. Affected value/scope: key/epoch-dependent evidence and paid exposure. Recovery authority: signed revocation/incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: acceptance outside declared compatible policy. Resource payer: adopting region and update carriers/archives; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R9. Owner key loss, theft and inheritance. Threat: irretrievable balances or colluding recovery guardians steal them. Assumption: secure custody and any pre-adopted guardian/observation trust. Mechanism: S12 explicit recovery roles, unique nonces and new-key binding; direct-key users accept irrecoverability. Residual: total key/guardian loss, coercion,

hidden compromise and indefinite absence defeat recovery guarantees. Acceptance: stolen-device, missing/compromised guardians, concurrent claims, delayed challenges and broken-old-key tests; no unauthorized reassignment or recovery from untrusted historical signatures. The user must acknowledge the chosen custody and inheritance tradeoff before real value.

R9 operational obligations. Detection/check: custody loss, authenticated guardian conflict and invalid claims. Affected value/scope: that owner policy and linked channels. Recovery authority: only guardians/roles adopted while ownership was trusted; exact identities, quorum and scopes must be signed before qualification. Stop rule: unsupported reassignment and unsafe challenges. Resource payer: owner and contracted guardian/watch services; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R10. Old backups and total local rollback. Threat: repeated signing, stale channel close, duplicate import or erased incidents. Assumption: a surviving qualified independent monotonic witness. Mechanism: S11 authentic head/floor/lock comparison and read-only recovery without proof of compatible freshness. Residual: loss or compromise of all witnesses blocks safe restoration. Acceptance: restore all files from an authentic older backup, clone custody to two devices, lose witness access and interrupt restore; signing/spending refuses until the independent current head is established. A same-directory head copy does not satisfy the test.

R10 operational obligations. Detection/check: missing/wrong independent head, floor, lock or incident commitment. Affected value/scope: restored custody and dependent available-value claims. Recovery authority: qualified independent witness and adopted incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: signing/spending until compatible freshness is proven. Resource payer: owner/region and independent witness services; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R11. Data extinction and physical disaster. Threat: all history, software, media or institutions disappear. Assumption: sufficient independent copies, funded renewal and surviving authentic witnesses. Mechanism: S16 original-byte archives, diverse failure domains, tool preservation and recoverability drills. Residual: correlated total loss or extinct institutions can permanently strand assets. Acceptance: recover after modeled regional/power/media/controller loss within stated scale and horizon; verify complete provenance, locks and tombstones, measure restoration cost and document which losses are unrecoverable. Logical aging is not a claim of actual million-year media survival.

R11 operational obligations. Detection/check: failed full recovery drill, damaged bytes and missing authentic witnesses. Affected value/scope: unrecoverable histories and dependent liabilities. Recovery authority: surviving pre-adopted archive/incident roles only; exact identities, quorum and scopes must be signed before qualification. Stop rule: new value from unavailable/untrusted history. Resource payer: funded custodians and adopting regions; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R12. Storage and cryptographic denial of service. Threat: adversarial input exhausts disk, replay/signature work, memory or connections. Assumption: admitted demand remains inside declared budgets and protected accepted obligations can be retained. Mechanism: S13/S14 canonical cheap screening, bounded authentication, quotas, backpressure and fair scheduling. Residual: a sufficiently large attack can deny availability; refusal cannot create funds or guarantee service. Acceptance: malformed maximum-size proofs, dependency-depth bombs, queue saturation, disk-full/fsync failure, floods and honest minority traffic; bound memory/CPU/disk/wait, preserve accepted custody and monetary records, and report both completed and censored requests. Sample peaks alone are not sustained resource qualification.

R12 operational obligations. Detection/check: quota/work/service limits, fsync failures and honest-request starvation. Affected value/scope: accepted custody and value requiring new work. Recovery authority: local declared admission authority; no monetary override; exact identities, quorum and scopes must be signed before qualification. Stop rule: new obligations before protected resources are exhausted. Resource payer: peer admission budgets, regional fees and operators; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R13. Long-term security funding and liquidity. Threat: fees/funding cannot pay validators, carriers, archives or renewals, while users demand guaranteed liquidity. Assumption: explicit voluntary service commitments and a budget adequate for declared scale. Mechanism: S14 separates origin subsidy, destination fees and carriage/archive escrow, with shortfall/stop rules; no hidden issuance. Residual: no mechanism guarantees future willingness, price or purchasing power. Acceptance: measured costs, stress/load/replay/renewal budgets, subsidy-to-fee transition scenarios, fee spikes, exhausted escrow and operator departure; liabilities conserve and new obligations halt at the policy's deficit threshold. Funding models need independent review; a supply cap is insufficient.

R13 operational obligations. Detection/check: measured funding deficits, unserved work and liquidity shortfall. Affected value/scope: service obligations and explicitly collateralized credit. Recovery authority: signed service/escrow and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: new obligations below adopted budget threshold. Resource payer: explicit fee beneficiaries, service customers and sponsors; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R14. Issuance rounding and overflow. Threat: platform-dependent rewards, cap overshoot or dust minted repeatedly. Assumption: exact signed Section 6 recurrence and checked integer semantics. Mechanism: era budgets, deterministic early-slot remainder allocation and reserve exhaustion. Residual: a defective implementation can violate arithmetic despite a correct written rule. Acceptance: cross-language/cross-platform vectors for genesis, heights 1/200000/200001, reserves 0/1/2, nondivisible budgets and overflowing input sets; every complete era releases exactly its budget and all remaining reserve reaches zero without exceeding 10^{35} . Replay rejects any divergent reward.

R14 operational obligations. Detection/check: independent recurrence replay and checked-arithmetic failures. Affected value/scope: invalid minted or consumed outputs. Recovery authority: adopted origin/finality and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: acceptance of any noncanonical amount/reward. Resource payer: origin operators and independent reviewers; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R15. Channel observation and offline double spending. Threat: stale close wins while watchers are absent, or two isolated recipients accept one copied bearer claim. Assumption: qualified local watcher/inclusion window for channels; no unlimited software-only offline finality. Mechanism: prefunded escrow, owner-authenticated state, unilateral exit/challenge and S18's explicit separate offline credit/hardware claims. Residual: unbounded local outage, compromised watchers/devices and counterparty default can cause losses outside the qualified assumptions [15, 22]. Acceptance: stale-state closure, censorship, absent watcher, reorganization and expiry boundaries; complete local settlement before long departure or disclose the chosen trust/limit. Offline receipts never masquerade as independently settled native currency.

R15 operational obligations. Detection/check: observed stale close, lost watcher coverage and conflicting receipts. Affected value/scope: funded channel/credit exposure. Recovery authority: adopted owner/watcher/challenge rules and disclosed creditor; exact identities, quorum and scopes must be signed before qualification. Stop rule: unqualified channel receipt/offline-finality promises. Resource payer: channel owners and contracted watchers/credit providers; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R16. Relay eclipse, false routes and malicious couriers. Threat: forged discovery, censorship, altered destination or discarded payload. Assumption: authenticated reachable neighbors, bounded budgets and eventual at least one useful honest carriage path for liveness. Mechanism: S2/S13 purpose-separated identities, pinned roles, signed path bindings, duplicate safety and alternative contacts. Residual: signatures do not make an advertised route real or deliver through all-malicious/no-contact paths. Acceptance: N1-N10 identity/path/replay/refusal/restart/asymmetric-contact cases, honest destination verification and a physically or locally measured alternative-path delivery; zero monetary authority from advertisements or custody receipts.

R16 operational obligations. Detection/check: invalid advertisements/path bindings, refused custody and contact divergence. Affected value/scope: carrier obligations and still-pending exports. Recovery authority: declared relay

contract and regional incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: false availability claims and unqualified new custody. Resource payer: sender/carrier contracts and relay sponsors; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R17. Confidentiality and metadata. Threat: public provenance identifies users, recorded traffic decrypts later or route metadata enables targeting. Assumption: baseline public ledger disclosure and finite transport-suite horizons. Mechanism: S15 minimal logs/key reuse, reviewed payload protection and explicit privacy labels. Residual: required public amounts/dependencies and historical disclosures remain; traffic analysis may persist under encryption. Acceptance: published data-flow/threat review, wrong-recipient/decryption/tampering tests, log scans excluding secrets and realistic metadata correlation analysis. Any stronger privacy proposal must prove conservation/incident compatibility and its resource cost first.

R17 operational obligations. Detection/check: data-flow/log audit, unauthorized access and correlation/decryption review. Affected value/scope: disclosed identity/payment metadata; not a new balance. Recovery authority: owner policy and adopted data/incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: unsupported confidentiality/anonymity claims. Resource payer: owners and privacy/transport service operators; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R18. Clock, entropy and hardware faults. Threat: bad randomness leaks keys, clock rollback expires/extends policy incorrectly, or torn writes release signatures without locks. Assumption: qualified entropy, durability and any clock used for local policy. Mechanism: explicit height/epoch domains, anti-rollback witness, atomic persistence and qualified clock uncertainty; stellar consensus never uses simultaneous wall time. Residual: undiscovered device failures or physical compromise remain. Acceptance: entropy-source refusal/reseed boundaries, wall-clock jumps, power-loss at every custody/ledger release boundary and hardware replacement; no unsafe signature, custody acknowledgment or spendability claim from uncertain state.

R18 operational obligations. Detection/check: clock/entropy/durability health refusal and inconsistent witnesses. Affected value/scope: device-authorized signatures and dependent state. Recovery authority: pre-adopted witness/custody and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: uncertain signing, custody acknowledgment and spending. Resource payer: device owners and regional custody operators; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R19. Software and supply-chain compromise. Threat: malicious dependencies/builds, parser differences or wallet intent substitution. Assumption: reviewed deterministic profile, independent verifiers and authenticated release adoption. Mechanism: S17 reproducibility, provenance, isolation, bounded parsing and independent adversarial review [38]. Residual: multiple implementations may share bugs or compromised compilers. Acceptance: independent rebuilds and differential malformed-input/state-transition campaigns, targeted fuzzing and signed upgrade/rollback refusal; exact artifacts and unresolved critical findings published. Zero known findings does not establish zero future bugs.

R19 operational obligations. Detection/check: differential disagreement, provenance failure and authenticated advisories. Affected value/scope: operations using compromised artifacts. Recovery authority: adopted release/upgrade and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: unsafe execution/value operations pending qualified patch. Resource payer: release maintainers, adopting operators and review sponsors; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R20. Misleading finality and financial claims. Threat: users spend a receipt, stale UI or advanced credit as if it were final native balance. Assumption: current independently verified local evidence and clearly disclosed profile/credit risk. Mechanism: S18 complete state separation and fresh pre-sign checks. Residual: social deception and mistaken trust remain possible. Acceptance: adversarial UI/node end-to-end tests for every stage, stale API, fabricated receipt,

mixed/quarantined balance and lost return link; display only proven state and refuse unsupported signing. No speed, liquidity, return or consumer-product claim without scoped evidence.

R20 operational obligations. Detection/check: wallet/node evidence mismatch and stale qualification. Affected value/scope: attempted payment and credit exposure. Recovery authority: owner-authorized wallet policy and adopted incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: unsupported signing and finality/return claims. Resource payer: wallet maintainers, owners and credit counterparties; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R21. In-flight upgrades and evidence renewal. Threat: a long journey outlives its authorization suite or an upgrade forgets old consumption. Assumption: still-authentic complete originals, surviving trust anchors and compatible local adoption within finite horizon. Mechanism: Section 12/I11 renewal before loss of authenticity, versioned identifiers and preserved permanent spent records. Residual: journeys longer than qualified cryptographic/institutional continuity may be inadmissible or stranded. Acceptance: multi-epoch exports, altered hash/address namespaces, retired suites, interrupted renewal and already-broken original evidence; no reinterpreted ID, resurrected balance or acceptance from unauthentic renewal. Stateful hash-based signature candidates additionally require qualified one-time-state management; they are not a backup-safe drop-in alternative [39].

R21 operational obligations. Detection/check: expiry, conflicting renewal or missing independent trust ancestry. Affected value/scope: in-flight and dormant old-suite liabilities. Recovery authority: pre-adopted suite/renewal and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: new acceptance from already-untrusted originals. Resource payer: regions, owners and renewable archive/carriage contracts; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R22. Invalid compression and missing data. Threat: a hash, snapshot, membership path or absence of a fraud proof is treated as full state validity. Assumption: the adopted full-replay requirement or a separately proved validity/data-availability profile. Mechanism: S7/S16/S17 reconstruct complete closure and retain original data; compact proofs bind exact rules and consumed/incident state. Residual: unavailable originals can prevent validation despite a valid commitment [20]. Acceptance: corrupt/truncated snapshots, erased ancestry, false roots, unavailable chunks and conflicting compact proofs; no credit from incomplete verification. Qualification includes full restoration cost and independent agreement.

R22 operational obligations. Detection/check: failed complete replay, unavailable data and invalid compact proofs. Affected value/scope: candidate credit and dependent history. Recovery authority: adopted verification/archive and incident roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: credit without required validity/data. Resource payer: verifying regions and contracted archives; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R23. Incident governance abuse and compensation. Threat: recovery authorities confiscate unrelated balances, suppress evidence or cover losses with hidden mint. Assumption: disclosed pre-adopted resolution roles and authentic liability accounting. Mechanism: S9/S10 provenance-scoped freeze, retained conflicts, public resolution and separate bounded collateral/credit claims. Residual: captured governance can still deny service or split adoption; already paid losses may exceed collateral. Acceptance: unauthorized/overbroad resolutions, duplicated compensation, loss of resolution quorum and reconnection of opposing decisions; reject invalid value changes, disclose shortfalls and keep affected value stopped. Legal redress is outside native consensus unless explicitly adopted.

R23 operational obligations. Detection/check: overbroad/unauthorized resolutions and inconsistent liability totals. Affected value/scope: disputed provenance and separately collateralized claims. Recovery authority: only pre-adopted incident/compensation roles; exact identities, quorum and scopes must be signed before qualification. Stop rule: invalid resolution and unsupported compensation. Resource payer: disclosed insurance/collateral/service sponsors, never hidden mint; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

R24. Claims beyond qualification and maintenance failure. Threat: a finite ground result is advertised as mainnet, independent stellar service or a hundred-million-year guarantee. Assumption: honest public qualification records and recurring independent reassessment. Mechanism: Section 20 gates, bounded claims, mandatory requalification and separately authenticated defect and implementation records. Residual: future unknown risks, discontinued civilization and dishonest marketing cannot be eliminated by this paper. Acceptance: an independent reviewer maps every public claim to exact code, custody, fault, capacity, horizon and route evidence; missing/failed/expired gates remain visibly unqualified. A whitepaper publication, build or activity log never satisfies protocol qualification.

R24 operational obligations. Detection/check: claim/evidence mismatch, expired qualification and new counterexamples. Affected value/scope: affected advertised services and unsupported acceptance. Recovery authority: adopted qualification/incident roles and owner-controlled document decisions; exact identities, quorum and scopes must be signed before qualification. Stop rule: out-of-scope claims and affected unsafe operations. Resource payer: operators, maintainers and independent review/renewal sponsors; no assumed unpaid eternal service. The threat above defines the adversary, and the acceptance case must exercise both detection and this stop rule.

20. Conformance and authenticated adoption

Foundational gates A-G. These definitions are complete within this paper; external project plans may track them but cannot change their obligations. Each gate requires reproducible exact artifact/identity bindings, failure reports and the applicable risk-register evidence. A gate's status is not inferred from the number of tests.

A. Authorization and identity. Verify the signed currency root and regional admissions, zero initial allocation, cap/units/recurrence, origin issuance authority, adopted rule/suite commitments and distinct roles. Rejected fixtures, reused keys, unauthorized roots/regions and substituted sources must never obtain real monetary authority. Independently reproduce adoption from authenticated public bytes.

B. Work, amounts and deterministic accounting. Recompute the reference profile's actual proof of work where applicable, or the target profile's authenticated proposal/finality eligibility; independently replay issuance, exact reward rounding, fees, maturity, payment, channels, imports, reorganization and recovery. Cross-platform implementations must agree on integer bounds and every state root, preserving I3/S8 without duplicate credit. Changing consensus does not waive accounting tests.

C. Nodes and participants. Produce reproducible bounded full-node packages, authenticate persistent synchronization, enable native default relay, recover after shutdown and actually operate with independent participants. Restrict administrative surfaces and show complete local payment progress when all remote endpoints are absent. Declare controllers, hardware, offered load and resource/fault budgets; operator telemetry is supplementary.

D. Release and adoption. Commit and independently rebuild exact source/artifacts, sign their immutable provenance, conduct separately held key/lock/witness recovery, and independently check actual node/wallet identities and rule floors. Test interrupted upgrades, rollback, malformed inputs and retirement. Qualification precedes fresh zero-allocation genesis adoption; a published website or test fixture is not a launch authorization.

E. Local payments and channels. Independent ordinary owners repeatedly send and receive mature locally verified value, pay explicit fees, use fully funded channels, exercise unilateral close/challenge, and recover custody under qualified local outages/reorganization and anti-rollback assumptions. Include mixed-input, group authorization, stolen/lost device and stale-display refusal. Measure completed and censored/failed payments against declared local latency/load only.

F. Asynchronous regional settlement. At least three actual native regions with distinct owner custody execute source debit/finality, evidence carriage, unique import/maturity, multiple-source split/merge, onward payment and return. Provide only neighboring contacts; remove origin services after export. Inject reordered/duplicated evidence, lost receipts, wrong destinations, partitions, signer conflicts, epoch rotation, self-certifying cycles and lineage quarantine. Check conservation at every transition and strict stopped replay. Preserve failures; no timeout refund or original-debit reuse.

G. Independent and sustained operation. Independent operators and reviewers reproduce security/custody/finality/recovery; qualify sustained adversarial load, growing history beyond the first 200,000 origin heights and stated production scale, archive/renewal budgets, physical loss, institutional succession and unresolved incident handling. Publish finite qualification horizons and stop/requalification triggers. Each real contact adapter and advertised physical route additionally requires measured operators, contact/capacity/trajectory evidence and fault recovery; a loopback campaign cannot qualify that route.

Native relay obligations N1-N10. These are mandatory under I8 and C/F/G, and do not depend on a mutable master plan. N1 requires persistent purpose-separated relay identity, authenticated network/region/version advertisements, tamper/wrong-network/unknown-version refusal and restart continuity. N2 requires ordinary full-node startup to enable discovery/relay without another process or mandatory Earth directory, with usable neighbor bootstrap and incremental reach demonstrated. N3 requires automatic bounded next-hop selection, directed forward/return routes, exact three-hop evidence delivery, alternate-path recovery and retention without a path. N4 requires durable store/carry/forward and restarted continuation under S13's custody policy; short handshake failures never erase monetary evidence or refund.

N5 requires every carriage boundary to authenticate exact payload, role, prior signed path and intended next hop, rejecting altered target, shortened signed path, loops and hop-bound violations. N6 requires duplicate/reorder idempotence, complete authenticated durable retention before custody acknowledgment/consumption, and capacity/failure atomicity preserving accepted and unprocessed evidence. N7 requires separate identity discovery, candidate route, observed contact, queued work, destination custody, ledger acceptance and maturity displays; advertisements and custody cannot claim online availability or spendability. N8 requires independent destination replay/admission/finality/unique-import validation; couriers acquire no issuance/mining/voting authority by connecting. N9 requires per-hop causal light time, directed contact plans, byte/capacity/backlog/trajectory/return analysis, fair scheduling and refusal under overload; no universal finite discovery or delivery deadline. N10 requires separately qualified cryptographic renewal, archives, independent operations and physical carriage horizons; seconds-long ground results cannot establish stellar or generational availability.

Specification and evidence package. Before qualification, publish an immutable complete protocol profile and model with: canonical message/state encodings and vectors; consensus/finality/view-change/reconfiguration rules and fault proof; governance/admission/incident/succession policies; S5/S6/S7/S8 transaction/provenance/accounting semantics; suites/epochs/renewal horizons; recovery witness and wallet/channel rules; limits and capacity/funding models. Attach independent implementations and review, reproducible campaigns and failures, exact source/binary/profile hashes, genesis/adoption/custody identities, hardware/contacts, configuration/seeds, offered load, per-stage budgets, independent operators, and retained unknowns. A composition proof must address safety for all permitted transitions; simulation and tests check implementation and find counterexamples. None alone proves all future risks absent.

Authenticated adoption requirements P1-P8. P1 selects and independently qualifies the exact consensus/epoch protocol and independent validator admission/fault model. P2 adopts separate governance thresholds, succession, incident and fork-resolution roles, with public accountability; an absent choice blocks affected authority. P3 selects current signature/hash/KEM profiles, overlap/retirement/renewal rules and finite horizons that cover admitted journeys. P4 selects direct-key versus guardian recovery, channel watcher assumptions and a qualified independent monotonic witness, explaining loss/collusion tradeoffs. P5 approves measurable resource, retention/archive and fee/funding budgets with deficit thresholds. P6 approves the complete canonical transaction/provenance profile, root/consumption retention and proof of arbitrary-region composition. P7 approves independently measured release/custody/security evidence, all A-G/I1-I12/N1-N10 gates and a fresh zero-allocation signed adoption excluding test assets. P8 authorizes only the specific physical routes, user/privacy claims and service horizons supported by evidence. Defaults do not constitute authenticated adoption; a missing policy or failed acceptance prohibits the affected authority. An adopted profile must satisfy these requirements rather than silently amend them.

Stop and requalification rules. Stop affected new value acceptance/signing when identity/profile authenticity, complete provenance, quorum/locks, independent freshness witness, suite horizon, incident consistency or protected resource/funding obligations cannot be established. Preserve records and allow read-only inspection and unrelated qualified activity. New code, changed critical dependencies/limits/custody, an exceeded fault/load/horizon bound,

compromise, incident, data loss or materially changed contact/physical service triggers a scoped new qualification. Reuse unchanged evidence only when exact bindings and assumptions remain valid. Requalification must retain failed evidence and satisfy the declared safety, maturity, quorum, resource and time bounds. Changing a bound defines a separately reviewed and authenticated profile; it cannot retroactively turn a failed claim into success, restore copied value or erase incidents.

21. Conclusion and protocol continuity

Protocol, algorithm, key, region and evidence versions remain mandatory. Authenticated evolution, local adoption, migration, witness/archive renewal and requalification follow Sections 11-12 and 18-20 while preserving ownership, conservation and finality. Implementations must retire unsafe cryptography through authenticated succession rather than continue operating under invalid assumptions. An upgrade cannot exceed the supply cap, erase liabilities or rewrite owner intent. Executable profiles are immutable and content-addressed at each adoption; successors must state compatibility and satisfy the same invariants. A proposal outside those invariants cannot claim conformance.

The hundred-million-year horizon is a purpose for successive humans maintaining locally useful, causally connected payment institutions. This paper gives a bounded conditional architecture, controls and explicit failure outcomes for that purpose. It cannot guarantee uninterrupted contact, civilization, purchasing power, authentic archives or secure mathematics throughout that time. Any operating claim must identify an exact implementation and route satisfying the published finite conditions, with maintenance and unresolved risks disclosed. Permanent loss or separation may leave value unavailable. No publication, future protocol or governance promise can honestly remove every risk.

References

- [1] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008. <https://bitcoin.org/bitcoin.pdf>
- [2] S. Burleigh et al., *Bundle Protocol Version 7*, RFC 9171, 2022. <https://www.rfc-editor.org/rfc/rfc9171.html>
- [3] K. Fall, *A Delay-Tolerant Network Architecture for Challenged Internets*, SIGCOMM, 2003. <https://people.eecs.berkeley.edu/~sylvia/papers/dtn.pdf>
- [4] J. Garay, A. Kiayias, and N. Leonardos, *The Bitcoin Backbone Protocol: Analysis and Applications*, EUROCRYPT, 2015. https://doi.org/10.1007/978-3-662-46803-6_10
- [5] C. Goes, *The Interblockchain Communication Protocol: An Overview*, 2020. <https://arxiv.org/abs/2006.15918>
- [6] CCSDS, *Schedule-Aware Bundle Routing*, 734.3-B-1, 2019. <https://ccsds.org/Pubs/734x3b1.pdf>
- [7] NASA, *Mars Communications Disruption and Delay*, 2023 Moon to Mars Architecture Concept Review. <https://www.nasa.gov/wp-content/uploads/2024/01/mars-communications-disruption-and-delay.pdf>
- [8] E. Birrane and K. McKeever, *Bundle Protocol Security (BPsec)*, RFC 9172, 2022. <https://www.rfc-editor.org/rfc/rfc9172.html>
- [9] M. Herlihy, *Atomic Cross-Chain Swaps*, 2018. <https://arxiv.org/abs/1801.09515>
- [10] M. Fischer, N. Lynch, and M. Paterson, *Impossibility of Distributed Consensus with One Faulty Process*, JACM, 1985. <https://groups.csail.mit.edu/tds/papers/Lynch/jacm85.pdf>
- [11] NIST, *Module-Lattice-Based Digital Signature Standard*, FIPS 204, 2024. <https://nvlpubs.nist.gov/nistpubs/fips/nist.fips.204.pdf>
- [12] C. Dwork, N. Lynch, and L. Stockmeyer, *Consensus in the Presence of Partial Synchrony*, JACM 35(2), 1988. <https://groups.csail.mit.edu/tds/papers/Lynch/jacm88.pdf>
- [13] R. Guerraoui, P. Kuznetsov, M. Monti, M. Pavlovic, and D.-A. Seredinschi, *The Consensus Number of a Cryptocurrency*, PODC, 2019; extended version. <https://arxiv.org/abs/1906.05574>
- [14] M. Yin, D. Malkhi, M. K. Reiter, G. G. Gueta, and I. Abraham, *HotStuff: BFT Consensus in the Lens of Blockchain*, PODC, 2019; extended version. <https://arxiv.org/abs/1803.05069>
- [15] A. Riard and G. Naumenko, *Time-Dilation Attacks on the Lightning Network*, arXiv preprint, 2020. <https://arxiv.org/abs/2006.01418>
- [16] J. Lind, O. Naor, F. Kelbert, I. Eyal, E. G. Sirer, and P. Pietzuch, *Teechain: A Secure Payment Network with Asynchronous Blockchain Access*, SOSp, 2019; extended version. <https://arxiv.org/abs/1707.05454>

- [17] Y. Hu et al., *A Delay-Tolerant Payment Scheme Based on the Ethereum Blockchain*, arXiv preprint, 2018.
<https://arxiv.org/abs/1801.10295>
- [18] J. E. Puente and C. Puente, *Bitcoin as an Interplanetary Monetary Standard with Proof-of-Transit Timestamping*, arXiv:2508.20591v1, 2025; preprint. <https://arxiv.org/html/2508.20591v1>
- [19] T. Alhajj and V. Corlay, *Improved Contact Graph Routing in Delay Tolerant Networks with Capacity and Buffer Constraints*, arXiv preprint, first posted 2024. <https://arxiv.org/abs/2410.15546>
- [20] M. Al-Bassam, A. Sonnino, and V. Buterin, *Fraud and Data Availability Proofs: Maximising Light Client Security and Scaling Blockchains with Dishonest Majorities*, arXiv preprint, first posted 2018. <https://arxiv.org/abs/1809.09044>
- [21] NIST, *Stateless Hash-Based Digital Signature Standard*, FIPS 205, 2024.
<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.205.pdf>
- [22] BIS Innovation Hub, *Project Polaris: Handbook for Offline Payments with CBDC*, 2023; institutional design handbook.
<https://www.bis.org/publications/project-polaris-handbook-offline-payments-cbdc>
- [23] J. Wang and H. Wang, *Monoxide: Scale Out Blockchains with Asynchronous Consensus Zones*, NSDI, 2019.
<https://www.usenix.org/conference/nsdi19/presentation/wang-jiaping>
- [24] NASA, *Our Nearest Celestial Neighbor? An Exotic 3-Star System*, 2025; astronomical distance reference.
<https://science.nasa.gov/exoplanets/other-stars-other-worlds/our-nearest-celestial-neighbor-an-exotic-3-star-system/>
- [25] NASA, *Voyager 1: What Is a Light-Day*, 2026.
<https://science.nasa.gov/mission/voyager/voyager-1/voyager-1-what-is-a-light-day/>
- [26] NIST, *Module-Lattice-Based Key-Encapsulation Mechanism Standard*, FIPS 203, final, 2024.
<https://csrc.nist.gov/pubs/fips/203/final>
- [27] NIST, *Recommendations for Key-Encapsulation Mechanisms*, SP 800-227, final, 2025.
<https://csrc.nist.gov/pubs/sp/800/227/final>
- [28] NIST, *Considerations for Achieving Crypto Agility: Strategies and Practices*, CSWP 39, updated edition, 2026.
<https://csrc.nist.gov/pubs/cswp/39/upd1/considerations-for-achieving-crypto-agility/final>
- [29] T. Gondrom, R. Brandner and U. Pordes, *Evidence Record Syntax (ERS)*, RFC 4998, 2007; archival-renewal reference, not ledger authority. <https://www.rfc-editor.org/rfc/rfc4998.html>
- [30] P. Shor, *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*, SIAM Journal on Computing, 1997; preprint. <https://arxiv.org/abs/quant-ph/9508027>
- [31] L. Grover, *A Fast Quantum Mechanical Algorithm for Database Search*, STOC, 1996; preprint.
<https://arxiv.org/abs/quant-ph/9605043>
- [32] G. Brassard, P. Hoyer and A. Tapp, *Quantum Cryptanalysis of Hash and Claw-Free Functions*, 1997; preprint.
<https://arxiv.org/abs/quant-ph/9705002>
- [33] S. Josefsson and I. Liusvaara, *Edwards-Curve Digital Signature Algorithm (EdDSA)*, RFC 8032, 2017.
<https://www.rfc-editor.org/rfc/rfc8032.html>
- [34] E. Rescorla, *The Transport Layer Security (TLS) Protocol Version 1.3*, RFC 8446, 2018.
<https://www.rfc-editor.org/rfc/rfc8446.html>
- [35] K. Kwiatkowski et al., *Post-Quantum Traditional (PQ/T) Hybrid Key Agreement Mechanisms for TLS 1.3*, RFC 10024, Standards Track, August 2026. <https://www.rfc-editor.org/rfc/rfc10024.html>
- [36] N. Bindel et al., *Hybrid Signature Spectrums*, RFC 9955, Informational, July 2026; terminology and analysis, not a concrete signature-combination standard. <https://www.rfc-editor.org/rfc/rfc9955.html>
- [37] J. Douceur, *The Sybil Attack*, IPTPS, 2002; identity/fault-independence boundary.
<https://www.microsoft.com/en-us/research/publication/the-sybil-attack/>
- [38] NIST, *Secure Software Development Framework (SSDF) Version 1.1: Recommendations for Mitigating the Risk of Software Vulnerabilities*, SP 800-218, final, 2022; development practices, not a protocol safety proof.
<https://csrc.nist.gov/pubs/sp/800/218/final>
- [39] NIST, *Recommendation for Stateful Hash-Based Signature Schemes*, SP 800-208, final, 2020.
<https://csrc.nist.gov/pubs/sp/800/208/final>